Screening Women Out?
Pay Transparency in Job Search

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Beyond employment: Where women work

- Women's education and employment is rising globally
- Yet, they stay clustered in low-wage, low-productivity firms
- Globally, women's sorting into low-wage firms $\Rightarrow$15-50% of pay gap (Card et al., 2016; Morchio and Moser, 2021; Bassier and Gautham, 2025)

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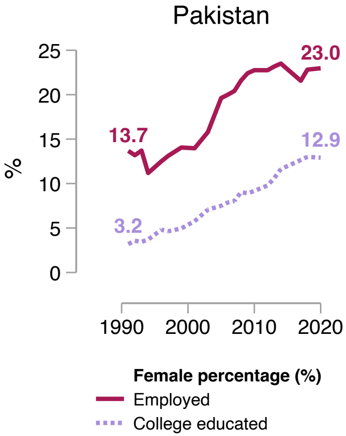

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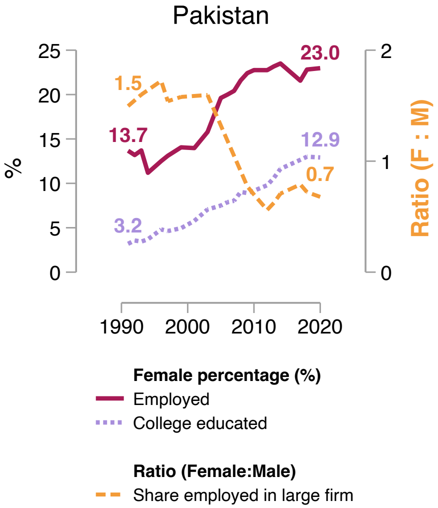

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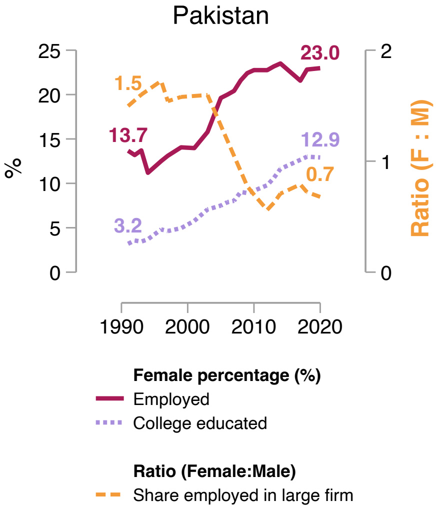
- Reflects preferences for "family-friendly" amenities? (Goldin, 2014; Pertold-Gebicka et al., 2016; Sorkin, 2017; Hotz et al., 2017; Kleven et al., 2019; Vattuone, 2023; Tô et al., 2025)



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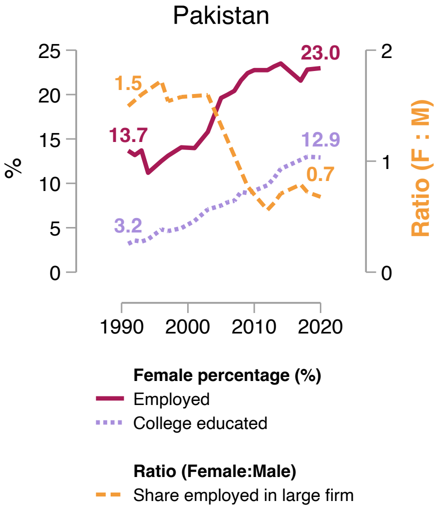
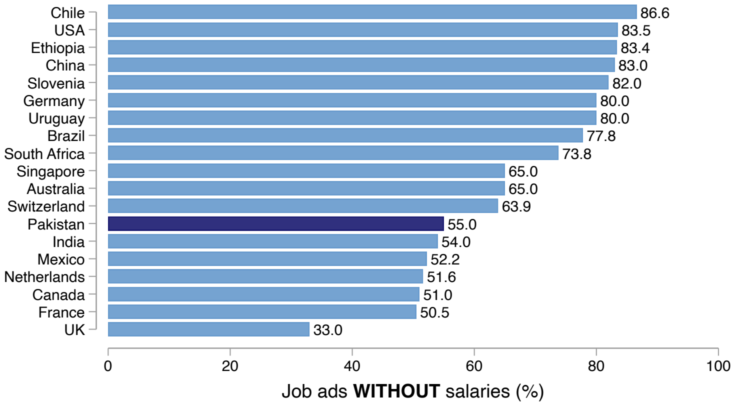
- Lab estimates of such preferences are small (Mas and Pallais, 2017)



Salary information is missing in most job ads



Details

This paper: Facts, theory and an experimental test to investigate sorting

Setting: Pakistan's largest online job search platform

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Equilibrium Facts

1. Admin data: 29M apps; 62K firms
2. Firm and worker surveys
3. Discrete choice experiment

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Equilibrium Facts ⇒ **Theoretical Framework**

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Theoretical Framework

1. Pay opacity magnifies modest amenity tastes
2. Gendered impacts

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Equilibrium Facts	⇒	Theoretical Framework	⇒	Field Experiment
1. Admin data: 29M apps; 62K firms 2. Firm and worker surveys 3. Discrete choice experiment		1. Pay opacity magnifies modest amenity tastes 2. Gendered impacts		1. Mandated salary disclosure 2. 20K jobs, 8900 firms, 310K workers 3. Saturation design for spillovers

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3. Discrete choice experiment				3. Saturation design for spillovers

The test: If wage information –

- **Has no effect on search:** Choices reflect non-wage preferences
- **Reallocates search:** Women lacked wage information, and do not value amenities more than pay

Findings: New explanation of gendered sorting and a policy solution

Large firm facts

1

3

Worker facts

2

4

Findings: New explanation of gendered sorting and a policy solution

Large firm facts	Worker facts
1 Higher pay, lower disclosure	2
3	4

Findings: New explanation of gendered sorting and a policy solution

Large firm facts	Worker facts
1 Higher pay, lower disclosure	2 Sort on pay; fail to infer when hidden
3	4

Findings: New explanation of gendered sorting and a policy solution

Large firm facts

1. Higher pay, lower disclosure
3. Offer less flexibility

Worker facts

2. Sort on pay; fail to infer when hidden
4. Women value flexibility more than men

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Transparency intervention nearly **doubles** women's applications to large firms, reverses gaps

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$\Rightarrow$ Gendered sorting occurs: Not because women prefer flexibility to pay
But because they don't know the price of flexibility

Roadmap

1. Experiment design
2. Search results
3. Mechanisms

Intervention: disable hiding feature for a random subset of jobs

20,088 jobs (8,906 firms) randomized into treatment or control

Control

Salary Range (Monthly)* From (PKR) To (PKR)

☐ Hide the salary from appearing on your job post.

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20,088 jobs (8,906 firms) randomized into treatment or control

Control

Salary Range (Monthly)*
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☐ Hide the salary from appearing on your job post.

Treatment

Salary Range (Monthly)*
From (PKR)
To (PKR)

To find you the best match, the salary for this job will be **displayed** in the ad, as part of a reform. Click here to learn more.

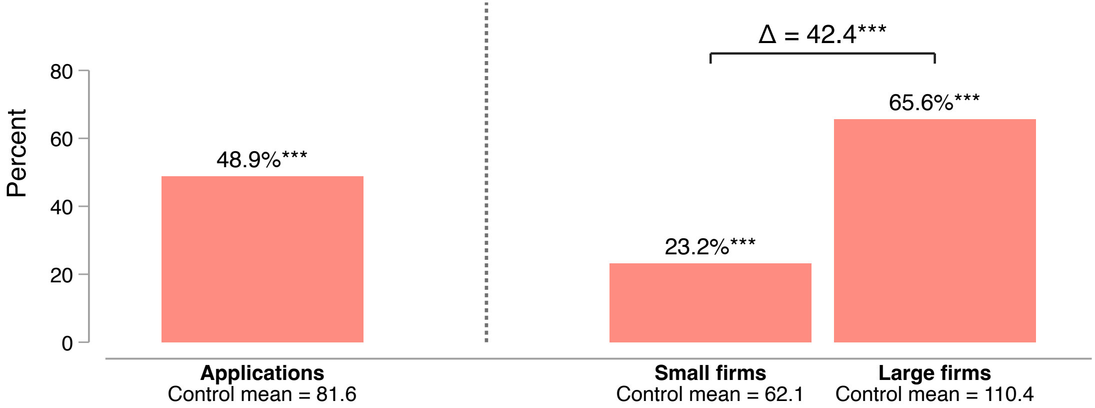
▸ Communication to firms ▸ Opt out mechanism ▸ Empirical approach ▸ Strong first stage ▸ Inattention check ▸ No spillovers ▸ No exits

Roadmap

Transparency raises applications

▸ OLS Table ▸ Poisson Table ▸ IV graph

Transparency raises applications, especially to large firms



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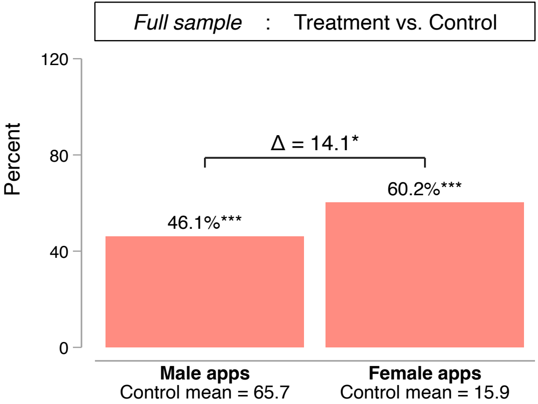
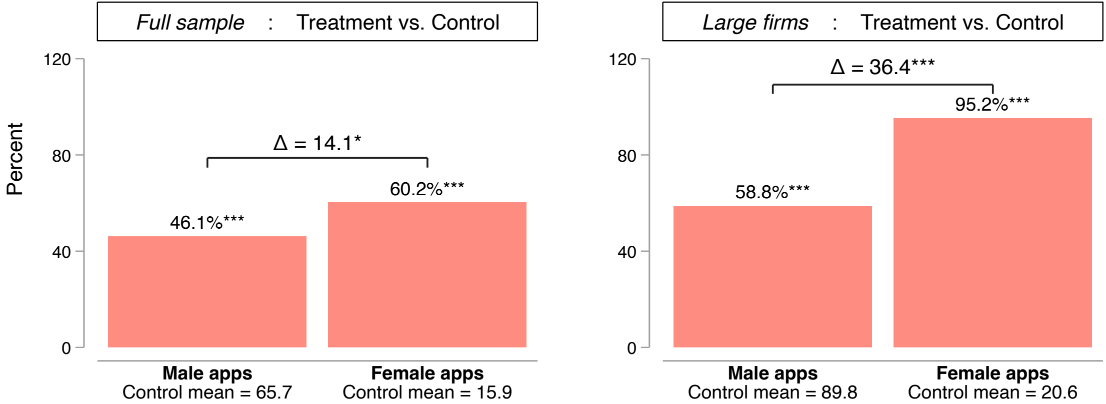
Transparency impacts larger on female apps



▸ OLS table ▸ Poisson table ▸ IV graph ▸ Worker level

Transparency impacts larger on female apps, especially to large firms



▸ OLS table ▸ Poisson table ▸ IV graph ▸ Worker level ▸ Gender gap in directed search

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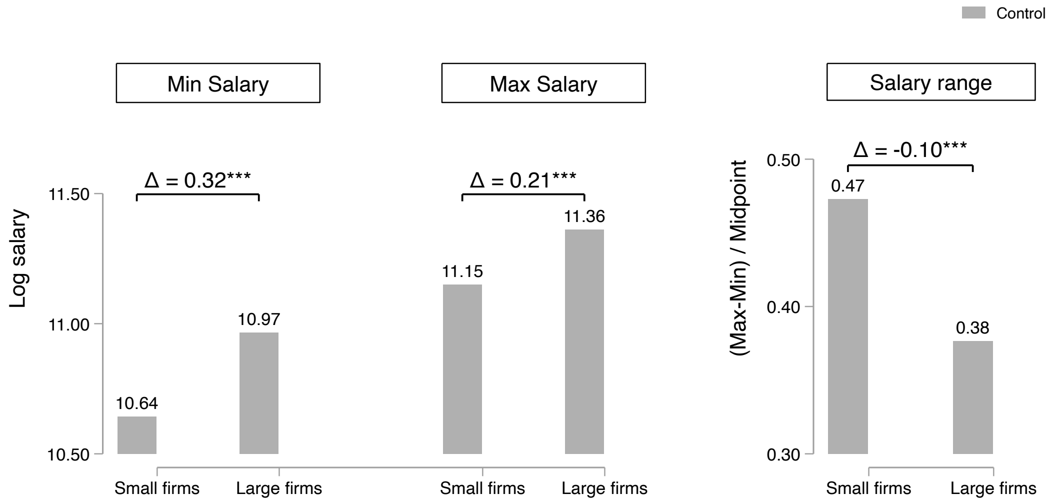
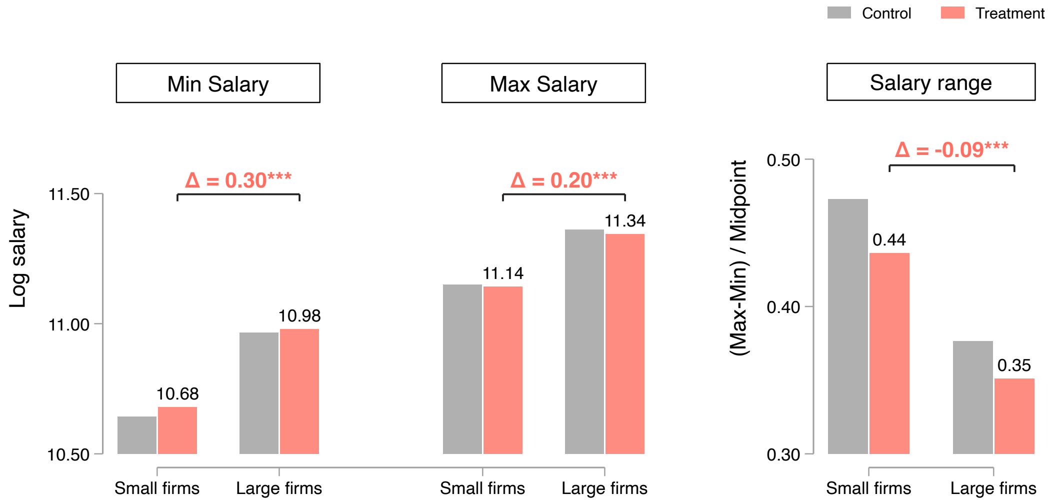
Large firms pay more than small firms in control



▸ Table ▸ IV ▸ Residualized ▸ Hidden salaries are higher

Large firms continue to pay more than small firms when treated



▸ OLS table ▸ IV table ▸ Residualized ▸ OLS HTE by firm size graph ▸ IV HTE by firm size graph ▸ Average TE graph

Large firms hide salaries more than small firms

▸ Treatment effect ▸ Baseline data

This paper: New explanation of gendered sorting and a policy solution

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Worker facts

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Salary non-disclosure binds for both genders

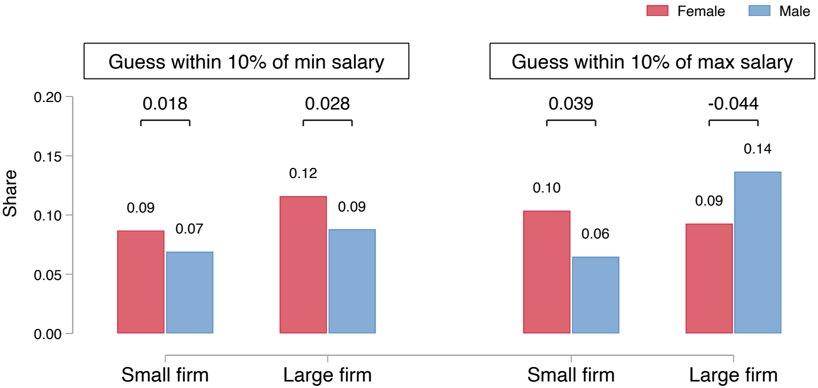
Survey of 464 workers (55% women) and incentivized guessing

- Only 11% of surveyed workers believe hidden salaries are higher than posted ones
  - 12.8% of men
  - 10.1% of women

Salary non-disclosure binds for both genders

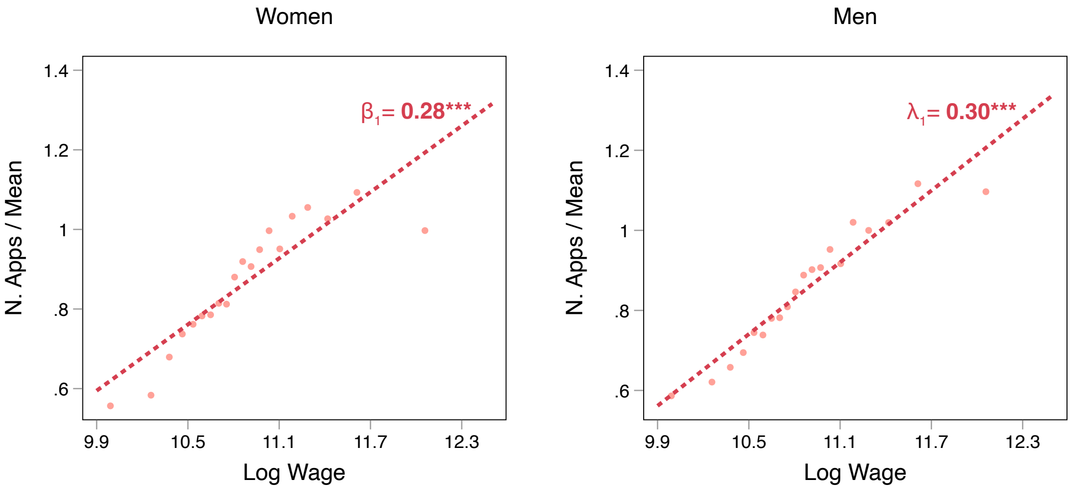
Survey of 464 workers (55% women) and incentivized guessing

- Only 11% of surveyed workers believe hidden salaries are higher than posted ones
- Lack of information about hidden salaries is not gendered



Men and women direct search on posted wages similarly

Baseline data on 29M apps to 300,000+ jobs 5 years pre-experiment



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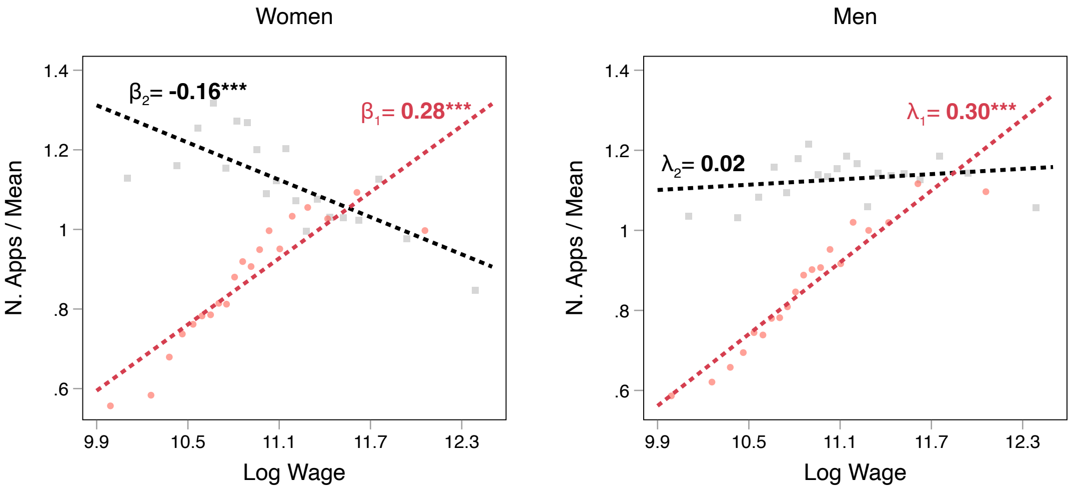

Test: $\beta_2 = \lambda_2$; p =0.00

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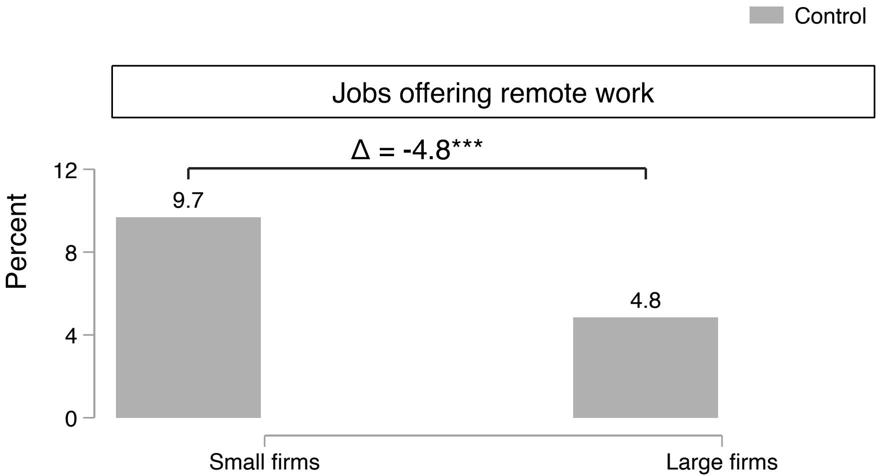
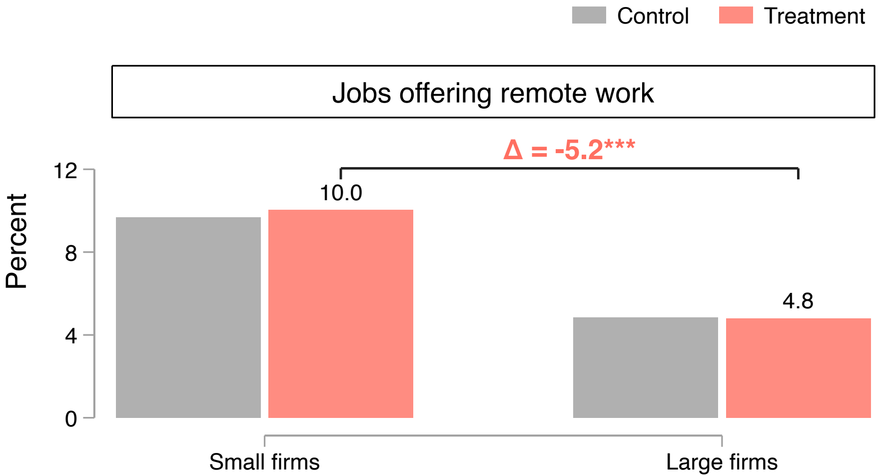
Large firms are less likely to offer remote work in control



▸ Baseline survey data

...remain less likely in treatment



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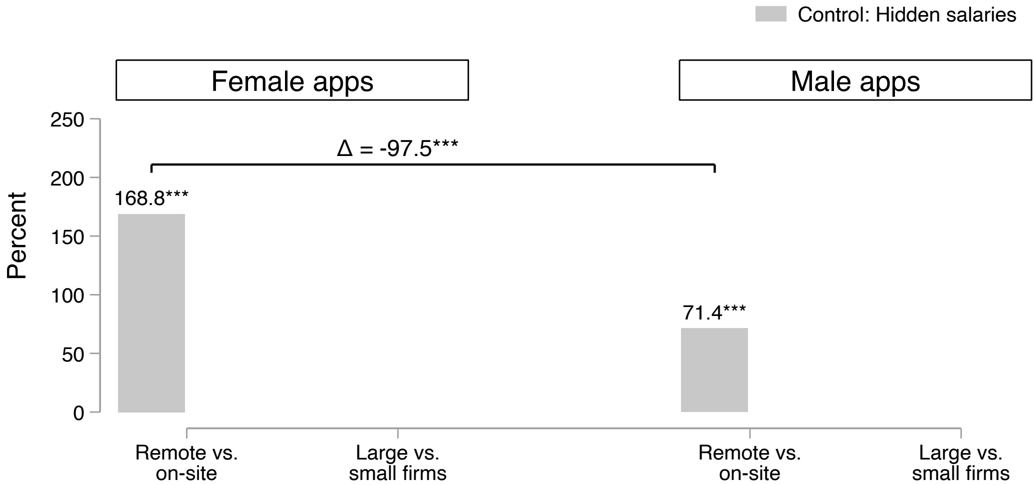
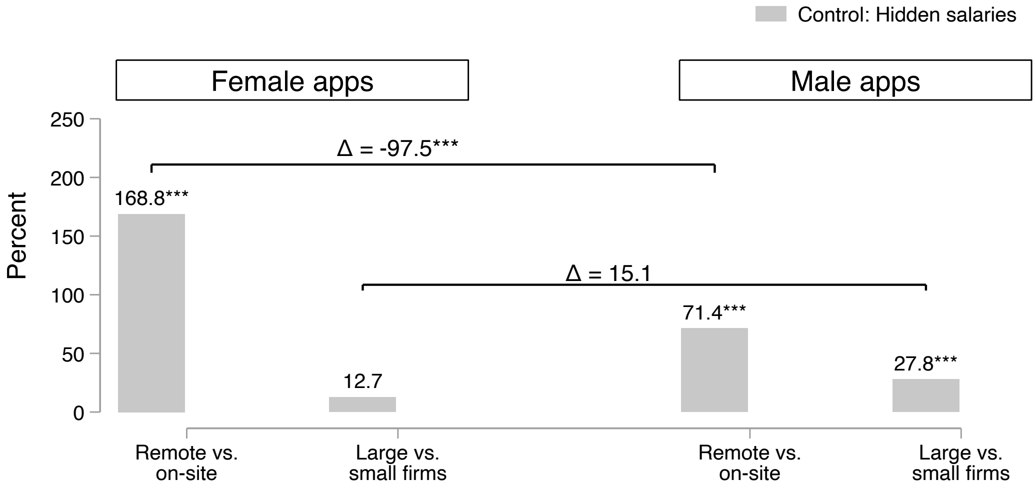
Remote jobs disproportionately draw women



▸ Lab: Women's moderate preference for flexibility ▸ Results replicated in the lab

Large firm jobs appeal to men but not women



▸ Lab: Women's moderate preference for flexibility

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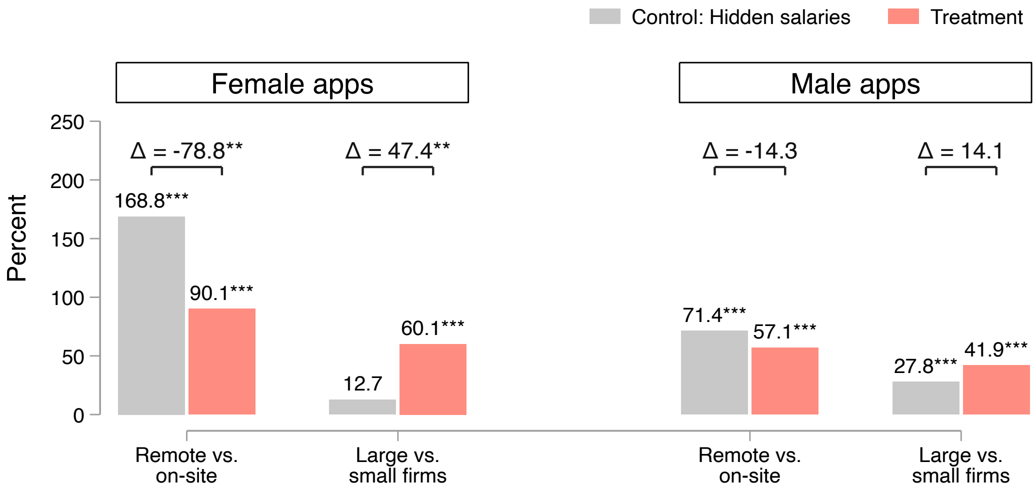
Remote work loses sizeable value for women when salary is disclosed



Lab: Women's moderate preference for flexibility | Results replicated in the lab | Similar pattern for other amenities

Alternative explanations

Gender gaps in beliefs about:

- Gender discrimination
- Competition
- Firm size

Gender gaps in preferences for:

- Pay transparency
- Higher wages
- Ambiguity aversion
- Bargaining aversion

Gender gaps in network composition

Conclusion

- Pay non-disclosure obscures the price of flexibility, directing women to low-paying firms
- Not discussed today:
  - Average quality declines but the best applicants in treatment vs. control are higher ability
  - More exposed firms $\Rightarrow$ more disclosure post-experiment
- Policies focused on revealing pay, rather than providing amenities, can go a long way at a low cost

References I

Arnold, David, Simon Quach, and Bledi Taska, "The Impact of Pay Transparency in Job Postings on the Labor Market," *SSRN Electronic Journal*, August 2022.

Banfi, Stefano and Benjamín Villena-Roldán, "Do High-Wage Jobs Attract More Applicants? Directed Search Evidence from the Online Labor Market," *Journal of Labor Economics*, 2019, *37* (3), 715–746.

Bassier, Ihsaan and Leila Gautham, "The firm-pay gender gap and formal sector churn over the life cycle," *Journal of Development Economics*, 2025, *176*, 103498. Short communication.

Batra, Honey, Amanda Michaud, and Simon Mongey, "Online Job Posts Contain Very Little Wage Information," Technical Report 31984, National Bureau of Economic Research December 2023.

Brenčič, Vera, "Wage posting: evidence from job ads," *Canadian Journal of Economics/Revue canadienne d'économique*, 2012, *45* (4), 1529–1559.

Card, David, Ana Rute Cardoso, and Patrick Kline, "Bargaining, Sorting, and the Gender Wage Gap: Quantifying the Impact of Firms on the Relative Pay of Women," *The Quarterly Journal of Economics*, 2016, *131* (2), 633–686.

Escudero, Verónica, Hannah Liepmann, and Damián Vergara, "Directed Search, Wages, and Non-wage Amenities: Evidence from an Online Job Board," Technical Report IZA Discussion Paper No. 17211, IZA — Institute of Labor Economics August 2024. Also issued as ILO Working Paper No. 136 (2025).

References II

Frimmel, Wolfgang, Bernhard Schmidpeter, Rene Wiesinger, and Rudolf Winter-Ebmer, "Mandatory wage posting, bargaining and the gender wage gap," Working Paper 2202, Linz 2022.

Goldin, Claudia, "A Grand Gender Convergence: Its Last Chapter," *American Economic Review*, 2014, *104* (4), 1091–1119.

Hotz, V. Joseph, Per Johansson, and Arizo Karimi, "Parenthood, Family Friendly Workplaces, and the Gender Gaps in Early Work Careers," NBER Working Paper 24173, National Bureau of Economic Research, Cambridge, MA December 2017. Revised June 2018; previously circulated as "Parenthood, Family Friendly Firms, and the Gender Gaps in Early Work Careers".

HRD Asia, "Pay transparency: More Singapore job postings including salary offers," June 2023. Accessed: 2024-04-15.

Indeed Hiring Lab, "Pay transparency: More Singapore job postings including salary offers," July 2023. Accessed: 2024-04-15.

__ , "Salaires dans les offres d'emploi : l'importance croissante de la transparence en Europe," April 2023. Accessed: 2024-04-15.

__ , "Salary Transparency Increasingly the Norm in Canadian Job Postings," March 2024. Accessed: 2024-04-15.

Jalal, Amen, "Screening women out? Experimenting with salary disclosure in job ads," *Working Paper*, 2024.

References III

— **and Kartik Srivastava**, “Do Application Caps Work?,” 2024.

Kleven, Henrik, Camille Landais, and Jakob Egholt Søgaard, “Children and Gender Inequality: Evidence from Denmark,” *American Economic Journal: Applied Economics*, 2019, *11* (4), 181–209.

Kuhn, Peter and Kailing Shen, “GENDER DISCRIMINATION IN JOB ADS: EVIDENCE FROM CHINA,” *The Quarterly Journal of Economics*, 2013, *128* (1), 287–336.

Mas, Alexandre and Amanda Pallais, “Valuing Alternative Work Arrangements,” *American Economic Review*, 2017, *107* (12), 3722–3759.

Morchio, Iacopo and Christian Moser, “The Gender Pay Gap: Micro Sources and Macro Consequences,” CEPR Discussion Paper DP16383, Centre for Economic Policy Research 2021.

People Matters, “Pay transparency on the rise in India, 50postings include salary details,” May 2023. Accessed: 2024-04-15.

Pertold-Gebicka, Barbara, Filip Pertold, and Nabanita Datta Gupta, “Employment Adjustments around Childbirth,” IZA Discussion Paper 9685, IZA Institute of Labor Economics January 2016.

Skoda, Samuel, “Directing Job Search in Practice: Mandating Pay Information in Job Ads,” *Job Market Paper*, November 2022.

References IV

Sorkin, Isaac, “The Role of Firms in Gender Earnings Inequality: Evidence from the United States,” *American Economic Review: Papers & Proceedings*, 2017, *107* (5), 384–387.

Tô, Maxime, É. Coudin, S. Maillard, and J. Tréguier, “Sorting into Lower-Paying Firms: The Firm Premium Child Penalty in the French Labour Market,” *Journal of Human Resources*, 2025. Revise & Resubmit.

Vattuone, Giulia, “Worker Sorting and the Gender Wage Gap,” 2023. Job Market Paper.

Roadmap

4. Appendix

Why do firms hide salaries?

- Baseline survey of 200+ firms who hide salaries ▸ Details

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  $\hookrightarrow$ Screening concerns
- 71% believe potential hires may be unworthy of the posted salary
  $\hookrightarrow$ Bargaining concerns
- But potentially over-estimate costs of transparency

	Firm ever hides salary post experiment		
	(1) **All firms**	(2) **Small firms**	(3) **Large firms**
Share of firm jobs treated	-0.111*	-0.075	-0.248**
	(0.059)	(0.068)	(0.114)
Control Job Mean	0.63	0.54	0.81
Observations	642	382	260

Gender difference in preference for remote work

◂ Back ▸ Design ◂ Model

Discrete-choice experiment with 438 workers

Job Posting 1

Pinnacle Partners is Hiring!

💼 Position: Software Engineer
📍 Location: Islamabad
🎓 Qualification: BA/BSc/B.Com/B.Ed
⏰ Job Type: Full-time (Monday to Friday, 9AM - 5PM)

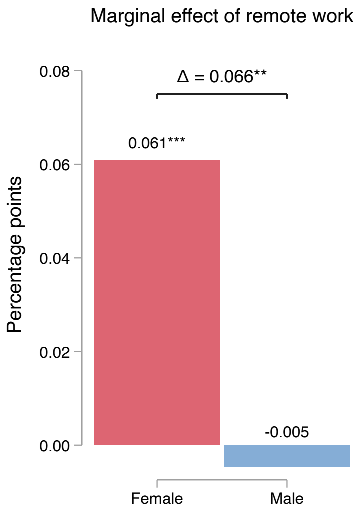
🏢 We are a **large** firm with over **100** highly-driven employees

Job Posting 2 SELECTED

Prime Innovations is Hiring!

💼 Position: Software Engineer
📍 Location: Islamabad
🎓 Qualification: BA/BSc/B.Com/B.Ed
⏰ Job Type: Full-time (Monday to Friday, 9AM - 5PM)

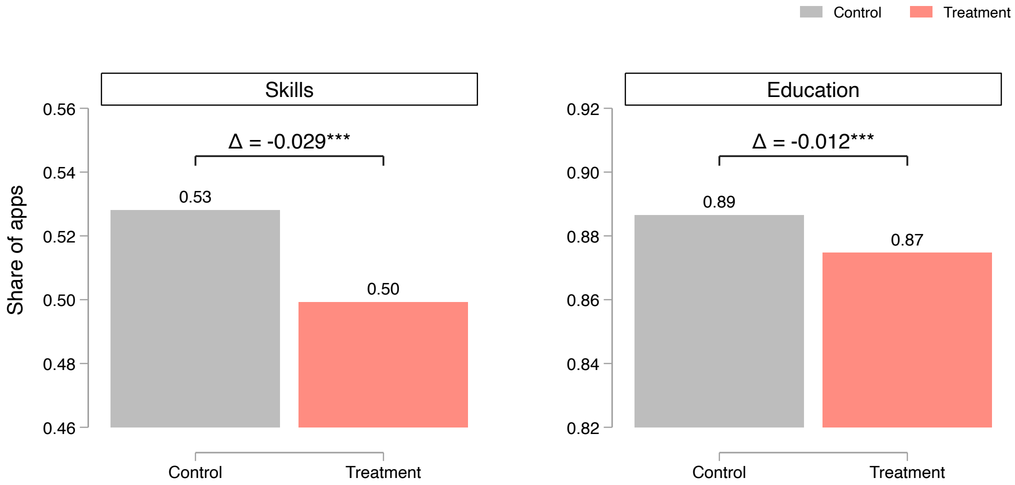
💰 Salary range: **PKR 63750 - PKR 93500**
🏠 This is a **remote** job. You will **work from home**
🌍 Our firm values **diversity**, offering **equal opportunities** to men and women



Modest decline in average quality among treated jobs

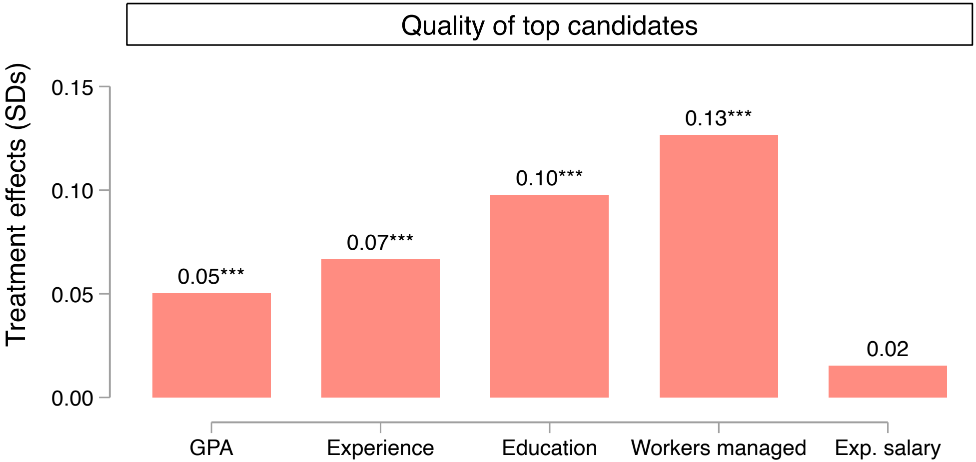
Decline larger for large firms

▸ Large Firms ▸ Small Firms



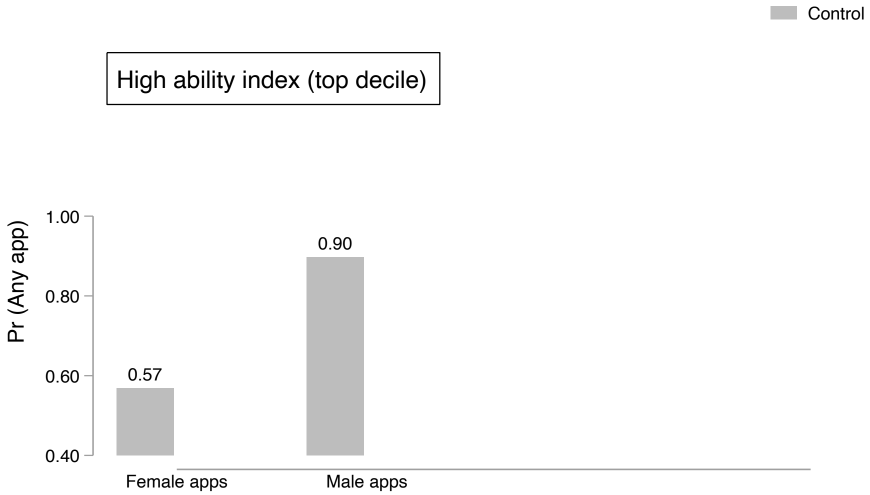
Top applicants in treated jobs are higher quality

▸ Large firms ▸ Small firms



▸ Construction

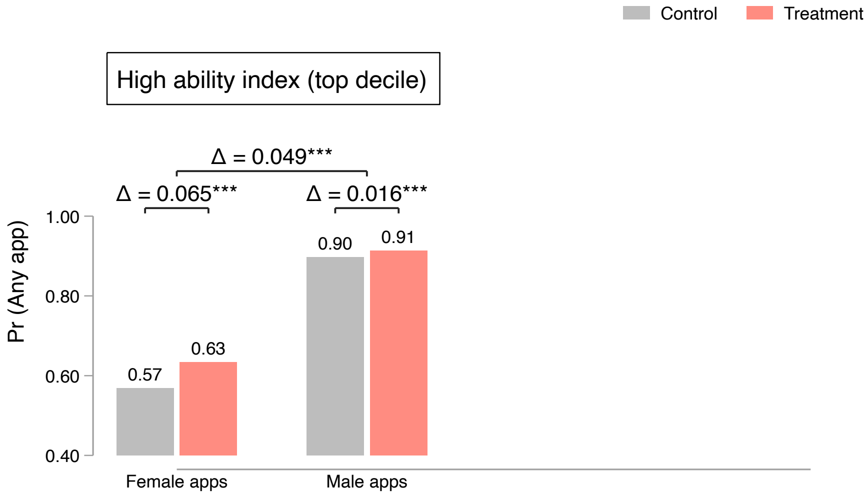
High likelihood that firms have high ability male applicants, but not female applicants



▸ Large Firms ▸ Small Firms

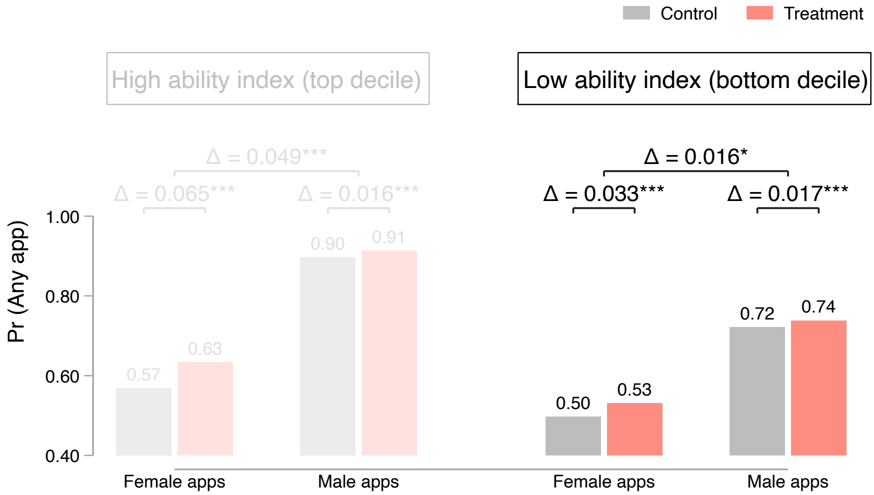
Treatment draws in high ability women

Gains smaller and gender difference insignificant for small firms



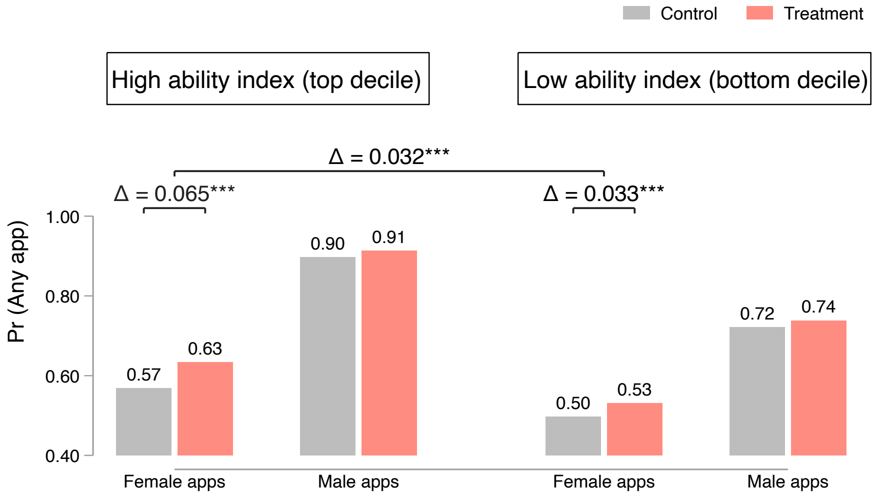
▸ Large Firms ▸ Small Firms

Trade off: more low ability applications



▸ Large Firms ▸ Small Firms

Relative increase largest for high ability women



▸ Large Firms ▸ Small Firms

Women's hiring likelihood at large firms: back of envelope calculation ▸ Details

May go up because more women apply, or down because there is more competition

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$$\Pr(\text{applicant hired}) = \Pr(\text{applies}) \times \Pr(\text{hired} \mid \text{applied})$$

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$$\text{Effect of treatment: } \frac{\Pr(\text{applicant hired} \mid \text{treatment})}{\Pr(\text{applicant hired} \mid \text{control})}$$

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$$\text{Effect of treatment: } \frac{\Pr(\text{applicant hired} \mid \text{treatment})}{\Pr(\text{applicant hired} \mid \text{control})}$$

- Assume random matching: a representative woman's chances of getting hired are $\frac{\text{N. women hired}}{\text{Total applications}}$
- **Firm survey:** Women equally likely to be hired in treated jobs despite more competition ▸ Table
- Since hiring chances are unchanged, hiring probability depends on each gender's application share

Women's hiring likelihood at large firms: back of envelope calculation ▸ Details

May go up because more women apply, or down because there is more competition

$$\Pr(\text{applicant hired}) = \Pr(\text{applies}) \times \Pr(\text{hired} \mid \text{applied})$$

$$\text{Effect of treatment: } \frac{\Pr(\text{applicant hired} \mid \text{treatment})}{\Pr(\text{applicant hired} \mid \text{control})}$$

- Assume random matching: a representative woman's chances of getting hired are $\frac{\text{N. women hired}}{\text{Total applications}}$
- **Firm survey:** Women equally likely to be hired in treated jobs despite more competition ▸ Table
- Since hiring chances are unchanged, hiring probability depends on each gender's application share

A representative woman is **17.8%** more likely to be hired at large firms in treatment than control (vs. -4.1% for men)

DCE: Remote work loses value for women when salary is disclosed

Back

Similar story with other amenities

	Control: Salary Hidden		Treatment	
	(1) N female apps	(2) N male apps	(3) N female apps	(4) N male apps
Large firm	1.13	1.28***	1.60***	1.42***
	(0.13)	(0.08)	(0.19)	(0.11)
Remote work	2.69***	1.71***	1.90***	1.57***
	(0.29)	(0.18)	(0.23)	(0.14)
Transport	1.18	1.02	0.56**	0.74**
	(0.24)	(0.21)	(0.13)	(0.11)
Flex work	1.03	0.89	0.66*	0.74***
	(0.16)	(0.11)	(0.14)	(0.08)
Safe environment	1.40	1.39*	1.02	1.01
	(0.31)	(0.27)	(0.17)	(0.12)
Equal opp. employer	1.73**	1.40**	1.27	1.24
	(0.39)	(0.22)	(0.31)	(0.19)
Dependent var. mean	18.31	72.07	25.52	95.98
Observations	3,851	3,851	10,972	10,972

Treatment effects accounting for neighbors' treatment

	N. Female Apps			N. Male Apps		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated Job	1.95*	-3.69***	-1.11	12.42***	-6.27**	-1.66
	(1.01)	(1.18)	(1.10)	(2.69)	(2.98)	(2.62)
Large firm	7.86***	7.20***	-3.11	40.48***	38.28***	0.71
	(1.87)	(1.82)	(2.37)	(4.37)	(4.26)	(5.33)
Treated Job × Large firm	17.67***	17.85***	13.05**	40.37***	40.96***	30.80**
	(5.87)	(5.88)	(6.48)	(13.33)	(13.35)	(14.60)
Constant	15.93	15.93	15.93	65.69	65.69	65.69
Observations	20,088	20,088	19,533	20,088	20,088	19,533
Control: high saturation cluster		✓			✓	
Cluster FE			✓			✓

Back

Control jobs by neighbors' treatment

	N. Female Apps		N. Male Apps		Salary visible
	(1)	(2)	(3)	(4)	(5)
High saturation cluster	1.31	-0.93	4.06	-0.38	-0.00
	(1.07)	(1.06)	(3.04)	(2.90)	(0.02)
Large firm		1.72		23.59***	
		(1.13)		(3.31)	
Constant	12.82	12.82	54.81	54.81	0.71
Observations	8,462	8,060	8,462	8,060	8,462
Cluster FE	✓		✓		✓
Drop 3 high traffic clusters in treatment		✓		✓	

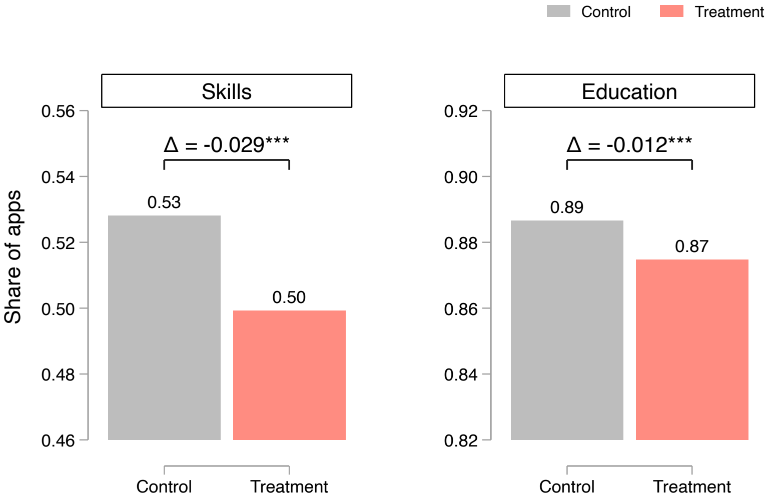
Back

Complementarities: treatment – high saturation

	N. Female Apps		N. Male Apps	
	(1)	(2)	(3)	(4)
Treated job	0.54	-0.32	3.56	2.09
	(0.90)	(1.02)	(2.62)	(3.14)
High saturation cluster	-1.74	-0.92	-3.90	-0.31
	(2.75)	(1.06)	(6.21)	(2.91)
Treated job × High saturation cluster	6.27	1.33	12.23	-0.29
	(4.26)	(1.28)	(9.59)	(3.81)
Constant	12.82	12.82	54.81	54.81
Observations	19,533	16,523	19,533	16,523
Cluster FE	✓		✓	
Drop 3 high traffic clusters in treatment		✓		✓

Back

LARGE FIRMS: Modest decline in average quality



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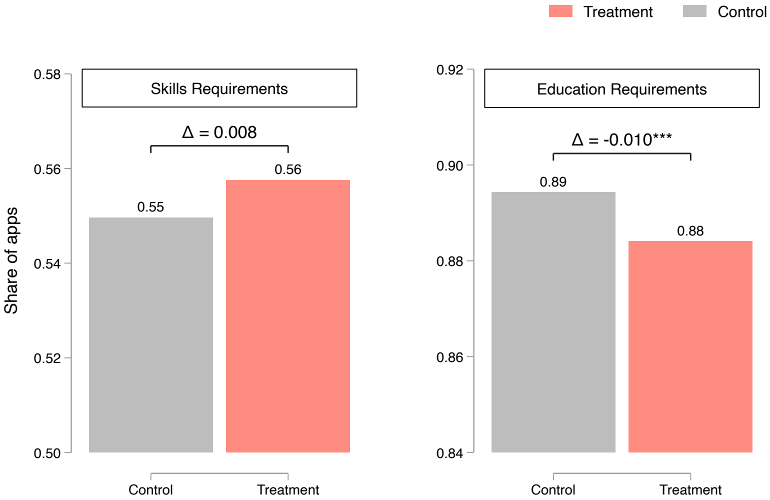
SMALL FIRMS: Modest decline in average education



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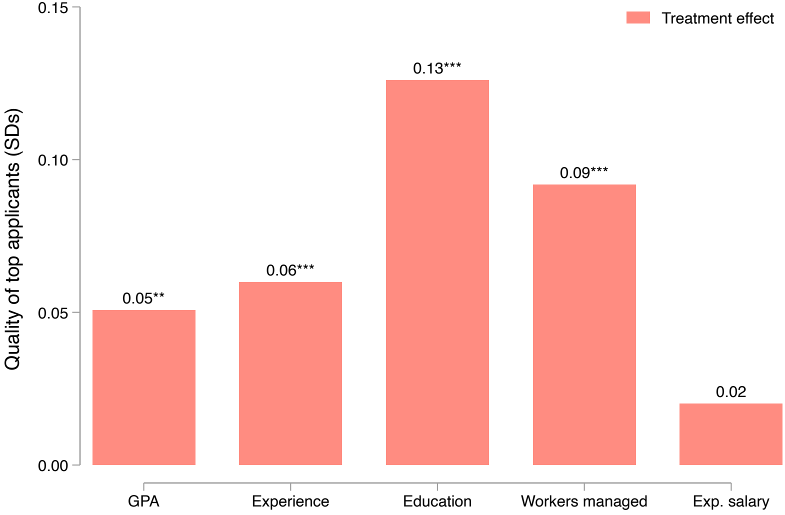
LARGE FIRMS: Top applicants are better quality

Back



SMALL FIRMS: Top applicants are better quality

Back

Views and applications to jobs with visible salaries increase

$\Longrightarrow$ Workers learn new, valuable information

	N views	N apps	% views	% apps
	(1)	(2)	(3)	(4)
Treatment	49.91***	39.91***	0.45***	0.49***
	(10.58)	(8.22)	(0.09)	(0.10)
Share of control mean	111.56	81.65	1.00	1.00
Observations	20,088	20,088	20,088	20,088

Back

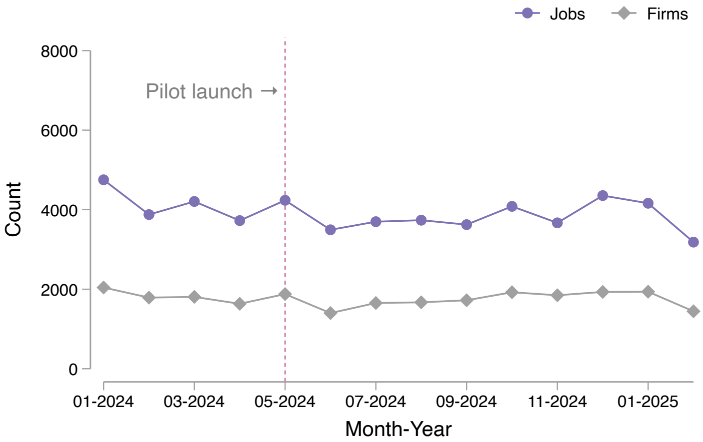
Views and applications to jobs with visible salaries increase

⟹ Workers learn new, valuable information

	N views	N apps	% views	% apps
	(1)	(2)	(3)	(4)
Treatment	49.91***	39.91***	0.45***	0.49***
	(10.58)	(8.22)	(0.09)	(0.10)
Share of control mean	111.56	81.65	1.00	1.00
Observations	20,088	20,088	20,088	20,088

Back

No decline in arrival of job posts at firm or job level



▸ Back to experiment design
▸ Back to first stage
▸ Back to conclusion

Large firms drive results (OLS)

	N views		N apps		% views		% apps	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	49.91***	18.24***	39.91***	14.38***	0.45***	0.16***	0.49***	0.18***
	(10.58)	(3.91)	(8.22)	(3.62)	(0.09)	(0.04)	(0.10)	(0.04)
Large firm=1		68.95***		48.37***		0.62***		0.59***
		(7.12)		(6.06)		(0.06)		(0.07)
Treatment × Large firm=1		71.63***		58.09***		0.64***		0.71***
		(24.68)		(19.09)		(0.22)		(0.23)
Control mean	111.56	83.66	81.65	62.08	1.00	0.75	1.00	0.76
Observations	20,088	20,088	20,088	20,088	20,088	20,088	20,088	20,088

Back

Large firms drive results (Poisson)

	N views		N apps	
	(1)	(2)	(3)	(4)
Treatment	1.45***	1.22***	1.49***	1.23***
	(0.10)	(0.05)	(0.11)	(0.07)
Large firm=1		1.82***		1.78***
		(0.11)		(0.12)
Treatment × Large firm=1		1.30**		1.34**
		(0.15)		(0.16)
Control mean	111.56	83.66	81.65	62.08
Observations	20,088	20,088	20,088	20,088

▸ Back

Large firms pay more; treatment does not change this

	(1) **Log Max Salary**	(2) **Log Min Salary**	(3) **Salary Range: (Max-Min)/Median**
Treated Job	-0.01	0.04***	-0.04***
	(0.01)	(0.01)	(0.01)
Large Firm	0.21***	0.32***	-0.10***
	(0.02)	(0.02)	(0.01)
Large Firm × Treated Job	-0.01	-0.02	0.01
	(0.02)	(0.02)	(0.01)
Constant	11.15	10.64	0.47
Observations	19,890	19,890	19,890

Back

Treatment effects by gender and firm size - OLS

***Panel A:** Levels*

	N male views		N male apps		N fem. views		N fem. apps	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	36.87***	15.02***	30.32***	12.43***	13.04***	3.23***	9.59***	1.95*
	(7.26)	(2.87)	(5.76)	(2.69)	(3.36)	(1.12)	(2.52)	(1.01)
Large firm=1		54.98***		40.51***		13.97***		7.86***
		(5.07)		(4.37)		(2.22)		(1.87)
Treatment × Large firm=1		49.11***		40.41***		22.53***		17.67***
		(16.91)		(13.34)		(7.87)		(5.87)
Control mean	86.31	64.07	65.72	49.32	25.24	19.59	15.93	12.75

***Panel B:** Percent change w.r.t. small firms in control group*

	% male views		% male apps		% fem. views		% fem. apps	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	0.43***	0.23***	0.46***	0.25***	0.52***	0.16***	0.60***	0.15*
	(0.08)	(0.04)	(0.09)	(0.05)	(0.13)	(0.06)	(0.16)	(0.08)
Large firm=1		0.86***		0.82***		0.71***		0.62***
		(0.08)		(0.09)		(0.11)		(0.15)
Treatment × Large firm=1		0.77***		0.82***		1.15***		1.39***
		(0.26)		(0.27)		(0.40)		(0.46)
Share of control mean	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Observations	20,088	20,088	20,088	20,088	20,088	20,088	20,088	20,088

Back

Treatment effects by gender and firm size - Poisson (Incidence Rate Ratios)

	N male views		N male apps		N fem. views		N fem. apps	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	1.43***	1.23***	1.46***	1.25***	1.52***	1.16***	1.60***	1.15*
	(0.09)	(0.05)	(0.09)	(0.07)	(0.14)	(0.06)	(0.17)	(0.09)
Large firm=1		1.86***		1.82***		1.71***		1.62***
		(0.10)		(0.11)		(0.13)		(0.17)
Treatment × Large firm=1		1.25**		1.27**		1.52***		1.69***
		(0.13)		(0.14)		(0.23)		(0.30)
Control mean	86.31	64.07	65.72	49.32	25.24	19.59	15.93	12.75
Observations	20,088	20,088	20,088	20,088	20,088	20,088	20,088	20,088

Back

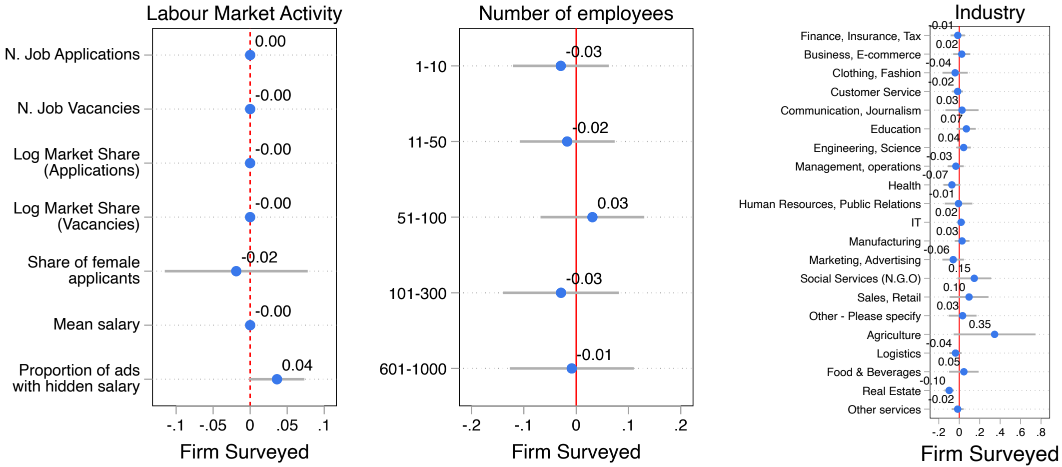
Treatment has little effect on compliance with skills desired

Share of Applicants Meeting All Desired Skills

	All Firms		Small Firms		Large Firms	
	(1) Male	(2) Female	(3) Male	(4) Female	(5) Male	(6) Female
Treated Job	-0.00	-0.00	0.00	0.00	-0.01	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Control Job Mean	0.30	0.30	0.31	0.32	0.28	0.29
Observations	21,619	20,111	12,692	11,812	8,927	8,299

Back

No imbalance on firm characteristics amongst respondents



F-test of joint significance: 1.37; p-val=0.07

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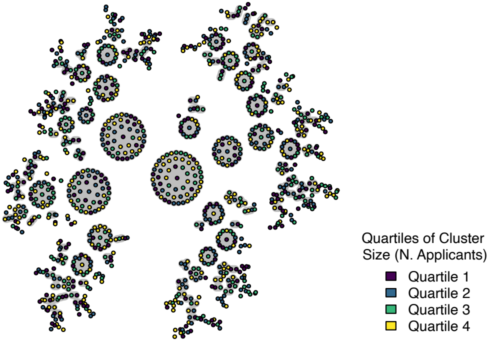
Two-staged randomization



Industry x Occupation clusters = 1118

Overlap adjusted Industry x Occupation clusters = 474

First stage: cluster randomization to measure spillovers

- Divided labor market into clusters of occupation $\times$ industry
- Adjusted for extent of overlap in applicants - grouped together clusters with more than 10% mutual overlap
- Randomized clusters into high (75%) and low (25%) treatment intensity

Second stage: randomize jobs into treatment

- Stratified to have about the same number of treated and control job ads

Back to experiment design Back to spillover results

Results are not driven by spillovers

▸ Details on two-stage randomization

Exploiting saturation design:

- No change in disclosure among control firms with more intensely treated competitors ▸ Table
- Control jobs receive similar applications regardless of treatment intensity among neighbors ▸ Table
- Treatment effects unaffected by treatment intensity of neighbors ▸ Table

Spillovers test: Conditioning on neighbors' exposure

	N. Female Apps			N. Male Apps		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated Job	1.95*	-3.69***	-1.11	12.42***	-6.27**	-1.66
	(1.01)	(1.18)	(1.10)	(2.69)	(2.98)	(2.62)
Large firm	7.86***	7.20***	-3.11	40.48***	38.28***	0.71
	(1.87)	(1.82)	(2.37)	(4.37)	(4.26)	(5.33)
Treated Job $\times$ Large firm	17.67***	17.85***	13.05**	40.37***	40.96***	30.80**
	(5.87)	(5.88)	(6.48)	(13.33)	(13.35)	(14.60)
Constant	15.93	15.93	15.93	65.69	65.69	65.69
Observations	20,088	20,088	19,533	20,088	20,088	19,533
Control: high saturation cluster		✓			✓	
Cluster FE			✓			✓

Back

Similar patterns documented in other contexts

- Batra et al. (2023) show in the U.S.
  - Wage information is difficult to access
  - Wages are more often disclosed in low income occupations
  - New in this paper:
    - Studying not just extensive margin presence of wages but the content of visible *and* hidden wages → able to show wages are higher when hidden for similar jobs
- Banfi and Villena-Roldán (2019) show in Chile
  - Job ads with visible wages require less experience, education and sector specific skills and have lower wages
  - But wages are “implicit” rather than being truly invisible in their setting - workers may use search filters by wage bracket
  - New in this paper:
    - Higher barriers to learning wages – more generalizable
    - Wage disclosure is more than a job level characteristic → shifting focus to firms: who hides and why?

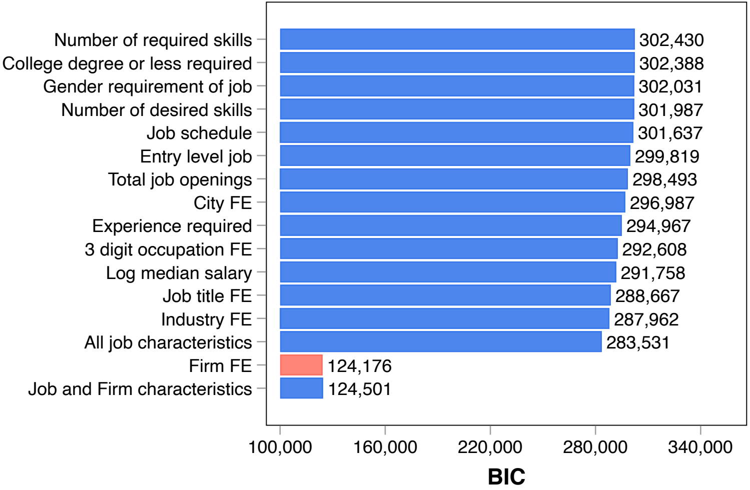
Back

Evaluations of policy

- Skoda (2022) studies effect of posting minimum salary in Slovenia. Finds
  - Wages of newly hired workers **increased** by about 3%
  - Job ads posted by firms induced to post salaries received **more clicks and applications** from job-seekers
  - **Negative selection**: applicants had on average lower pay expectations and characteristics less fitting job ads requirements
- Frimmel et al. (2022) studies effect of posting minimum salary in Austria. Finds
  - Larger reduction in gender pay gap for jobs which are immediately available and need to be filled urgently
- Arnold et al. (2022) studies spillover effects of disclosure of salary range. Shows
  - 30% non compliance
  - Posted wages **increased** by 3.6 ppt

Back

Firm FE minimize BIC



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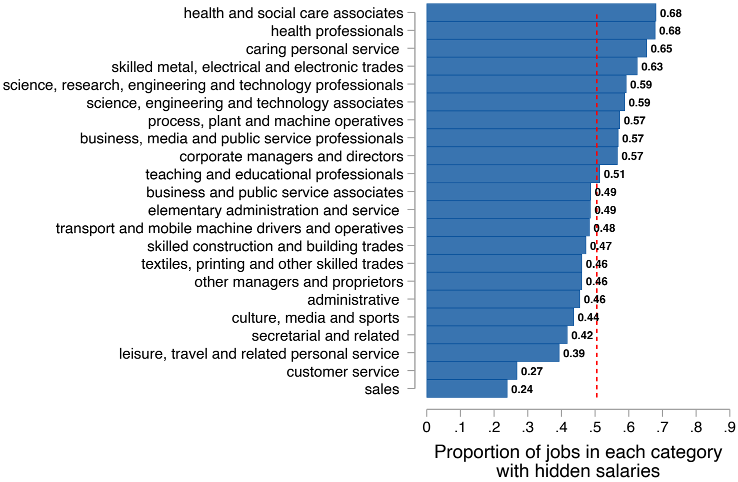
Substantial variation in salary hiding within occupation



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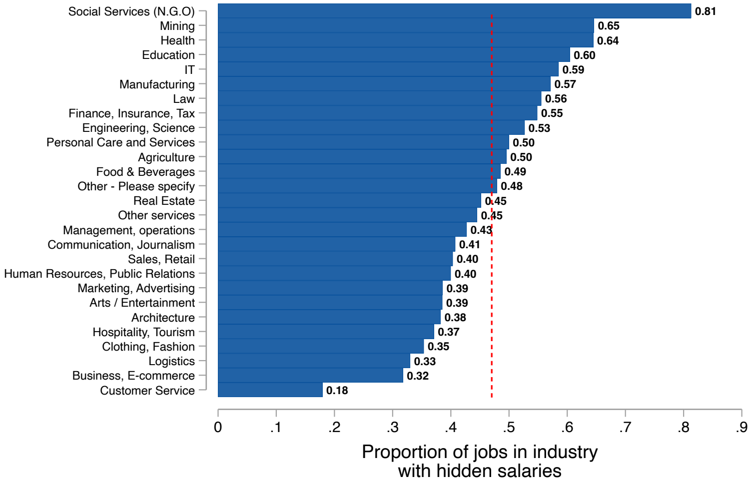
Substantial variation in salary hiding within industry



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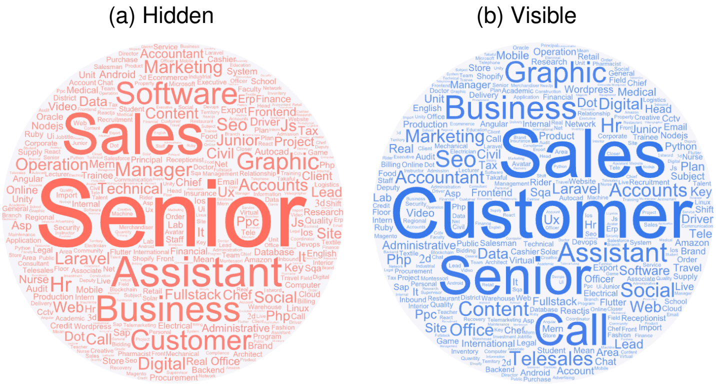
Considerable overlap in job titles of ads with hidden vs. displayed salaries

First Word of Job Title



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Job ads with hidden salaries

Continent	Country	Year	Percentage of job ads	Platform/Data source	Reference
North America	Canada	2024	51%	Indeed	Indeed Hiring Lab (2024)
	US	2012-17	80%-86.5%	Burning Glass Technologies (BGT)	Banfi and Villena-Roldán (2019), Batra et al. (2023)
South America	Chile	2008-14	86.60%	Trabajando.com	Banfi and Villena-Roldán (2019)
	Uruguay	2024	80%	BuscoJobs	Escudero et al. (2024)
Europe	France	2023	50.50%	Indeed	Indeed Hiring Lab (2023b)
	Germany	2023	80%	Indeed	Indeed Hiring Lab (2023b)
	Netherlands	2023	51.60%	Indeed	Indeed Hiring Lab (2023b)
	Slovakia	2019-22	77.2% (Before reform), 16.8% (After reform)	Profesia.sk	Skoda (2022)
	Slovenia	2001	82%	Local public employment agency	Brenčič (2012)
	UK	2023	33%	Indeed	Indeed Hiring Lab (2023b)
Asia	China	2008-10	83%	zhaopin.com	Kuhn and Shen (2013)
	India	2020-21	54%	Indeed	People Matters (2023)
	Pakistan	2019-24	55%	Rozee.pk	Jalal (2024)
	Singapore	2022	65%	Indeed	HRD Asia (2023)
Africa	Ethiopia	2023-24	83.38%	Afriwork	Jalal and Srivastava (2024)
Australia	Australia	2019-23	65%	Indeed	Indeed Hiring Lab (2023a)

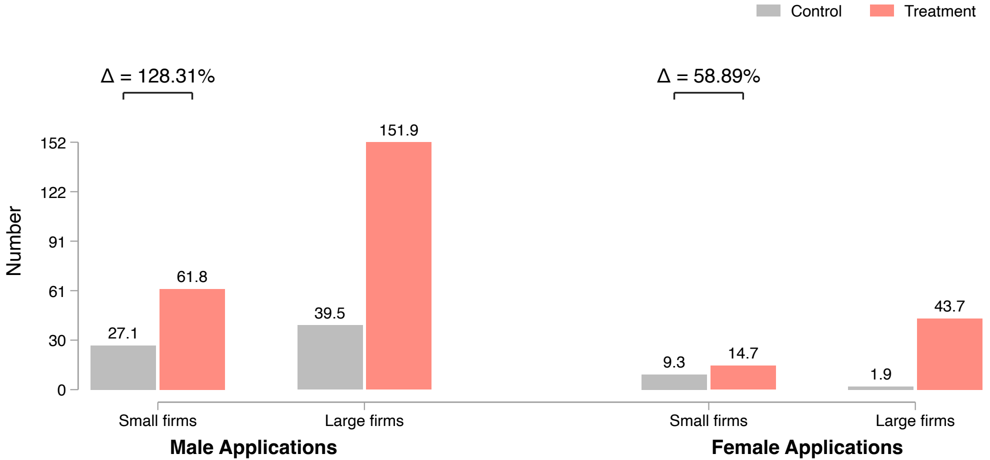
Back

First stage

	Visible Salary	
	(1)	(2)
Treatment	0.401***	0.358***
	(0.006)	(0.007)
Large Firm		-0.192***
		(0.011)
Large Firm $\times$ Treated Job		0.112***
		(0.011)
Constant	0.562	0.640
Non-compliance rate in treatment	0.037	
Non-compliance rate in small firms		0.003
Non-compliance rate in large firms		0.083
Observations	20,088	20,088

Back to graph

IV: Treatment has limited effect on applications to jobs from small firms



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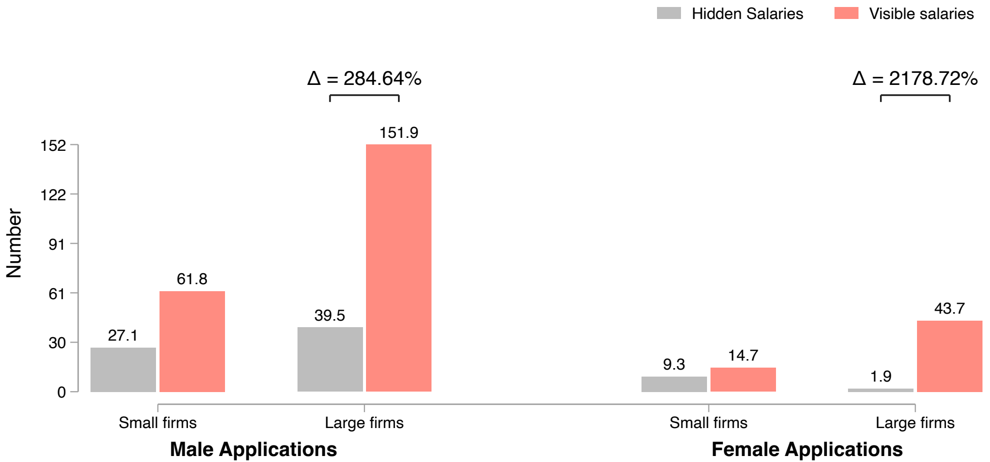
IV: Magnitude of reallocation from small to large firms is bigger for women



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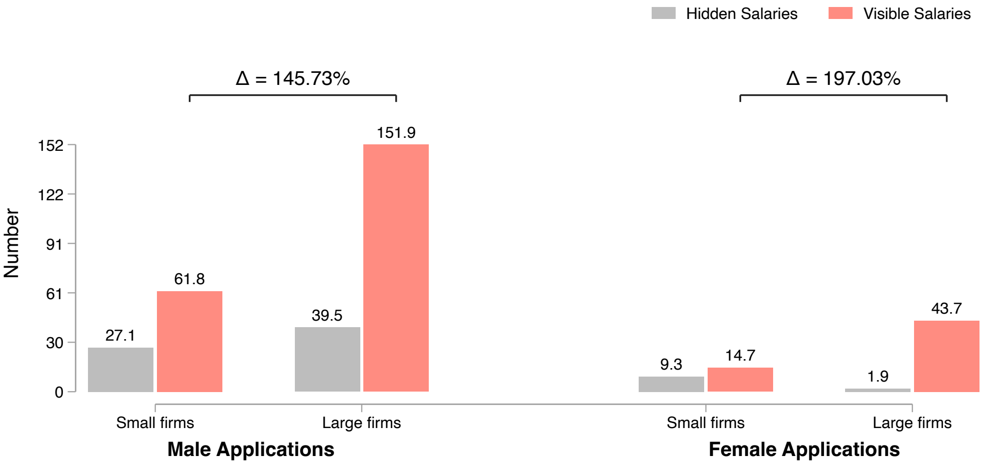
IV: Equilibrium gender gaps in allocation closed



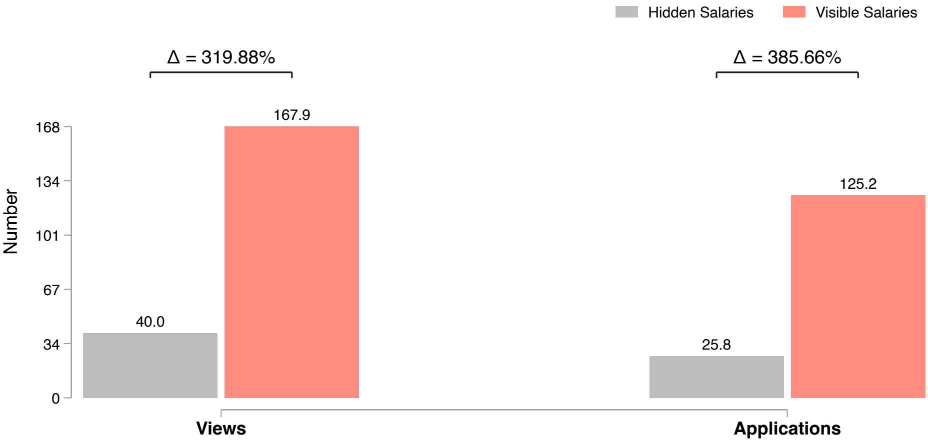
▸ Back

IV: Modest increase in min salary; range becomes more compact

▸ Back

IV: Views and applications to jobs with visible salaries increase

$\Longrightarrow$ Workers learn new, valuable information



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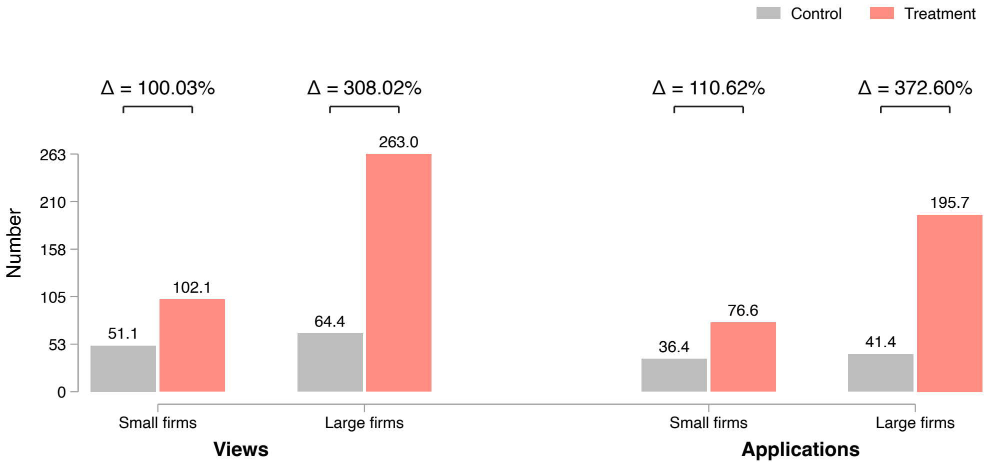
IV: Increase in views and apps driven by treated jobs at large firms



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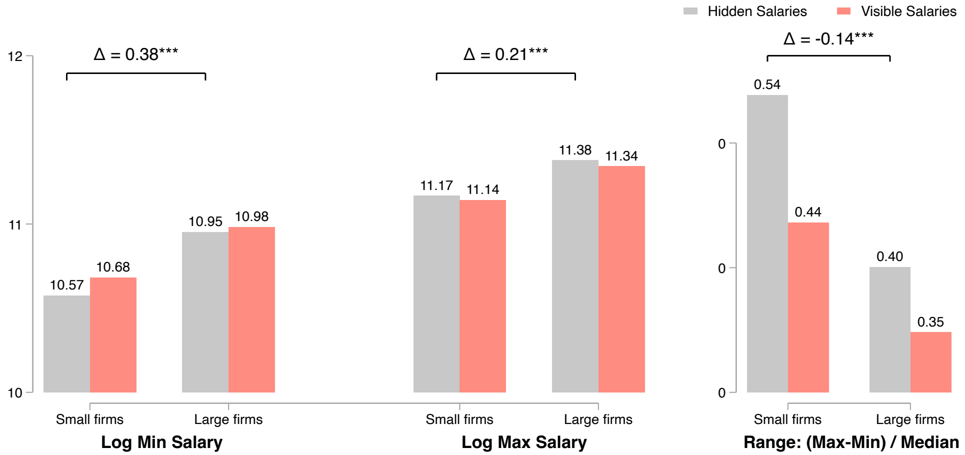
They pay more



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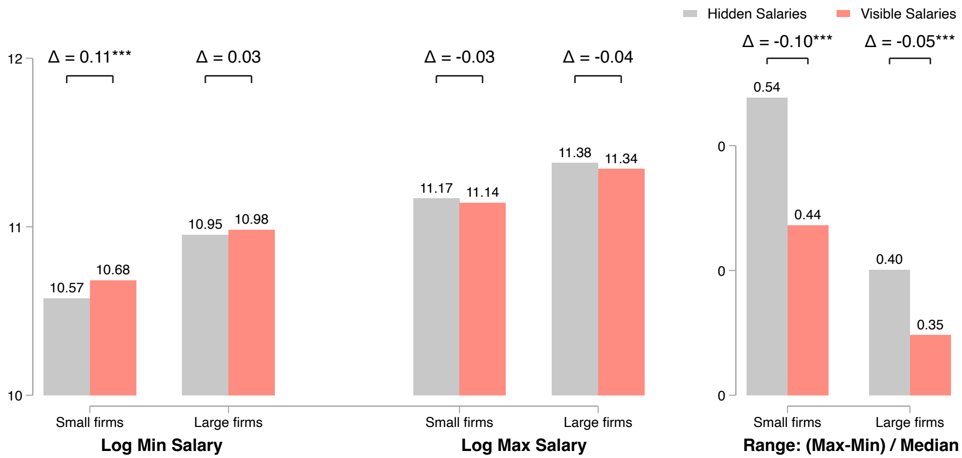
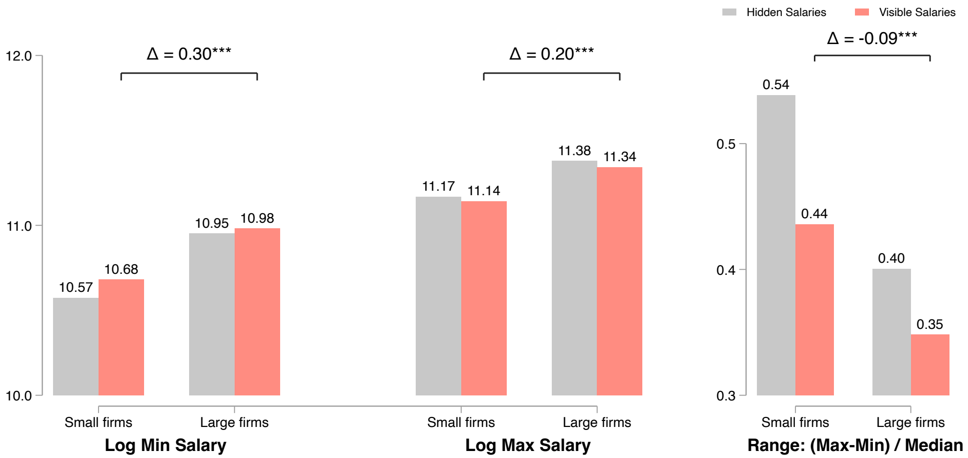
Treatment does not alter advertised salaries



▸ Back to OLS version ▸ Back to main results

They continue to pay more when treated



▸ Back

Inattention check

Confirmation ✖

Please be advised that the salary for your posted job will be displayed in the job details for potential candidates.

Confirm

▸ Back

Managing firm perceptions about the study: Purpose of the study

LEARN MORE

We anticipate that pay transparency in ads will enhance our ability to match you to the most suitable workers. Thus, we are implementing a reform in which salary ranges for select positions will be disclosed to jobseekers over the next five months. During this period, newly posted job ads will be randomly selected to display salary ranges.

Upon completion of this initiative, we eagerly look forward to sharing with you our insights on the effects of disclosing salary ranges in job ads, both on the market broadly, and on your firm specifically. In particular, we will share a detailed report with you tailored to your firm, on how your salary ranges compare to salaries in the market, and the quantity and quality of applicants you would have received for your specific job ad if its salary disclosure status were reversed. We will be supported in this analysis by economists from the London School of Economics. If you have any inquiries regarding this initiative, or if there is anything else you would like to learn from us, please feel free to reach out to us via email at [redacted] or on the number [redacted]. You can also contact LSE economists at **u.jalal@lse.ac.uk.**

Back

Managing firm perceptions about the study: Transparency about ongoing experiment

LEARN MORE

We anticipate that pay transparency in ads will enhance our ability to match you to the most suitable workers. Thus, we are implementing a reform in which salary ranges for select positions will be disclosed to jobseekers over the next five months. During this period, newly posted job ads will be randomly selected to display salary ranges.

Upon completion of this initiative, we eagerly look forward to sharing with you our insights on the effects of disclosing salary ranges in job ads, both on the market broadly, and on your firm specifically. In particular, we will share a detailed report with you tailored to your firm, on how your salary ranges compare to salaries in the market, and the quantity and quality of applicants you would have received for your specific job ad if its salary disclosure status were reversed. We will be supported in this analysis by economists from the London School of Economics. If you have any inquiries regarding this initiative, or if there is anything else you would like to learn from us, please feel free to reach out to us via email at [redacted] or on the number [redacted]. You can also contact LSE economists at **u.jalal@lse.ac.uk.**

Back

Opt-out mechanism - Non-compliance is observed

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Back

Use of online dashboard not differential by treatment

	(1) **CV views missing**	(2) **CV downloads missing**
Large Firm × Treated Job	0.02	-0.01
	(0.01)	(0.01)
Treated Job	-0.01*	-0.01
	(0.01)	(0.01)
Large Firm	-0.06***	-0.15***
	(0.01)	(0.01)
Constant	0.75	0.71
Observations	21,006	21,006

Back

Firm survey data

- 300+ firms surveyed in Dec. 2023
- Sampling:
  - Posted at least one ad in 2023; at least 5 ads during 2019-22
  - Stratified on hiding frequency to get a mix of salary hiding and non-hiding firms

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  - 81% have authority to change job ads on their own
  - 90% participate in setting salaries
  - Median respondent has 6 years of hiring experience
  - 32% were from HR, 41% from general administration (e.g., CEOs)

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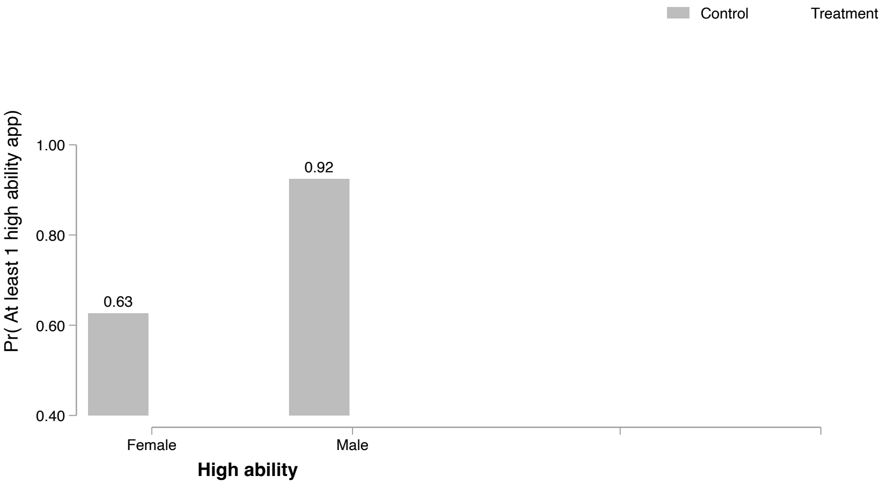
- Low response rate (15%) but no systematic selection into survey Balance tests

Back

LARGE FIRMS: Treatment most effective at drawing in high ability women



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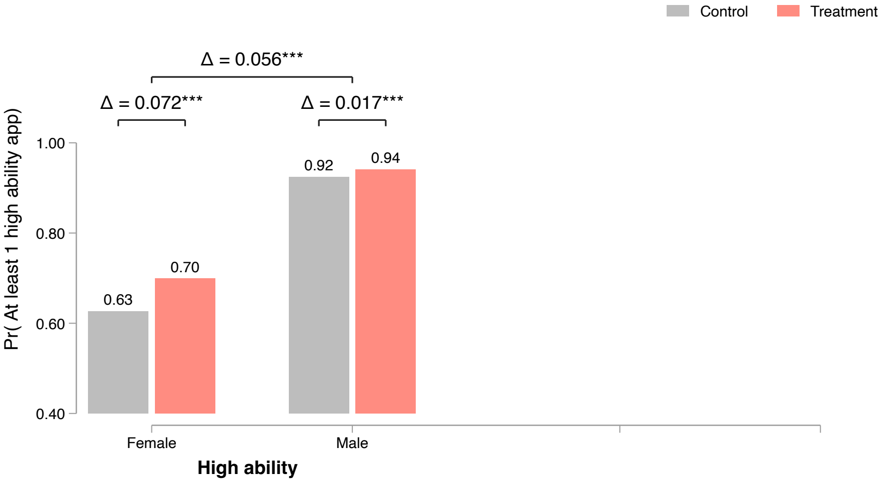

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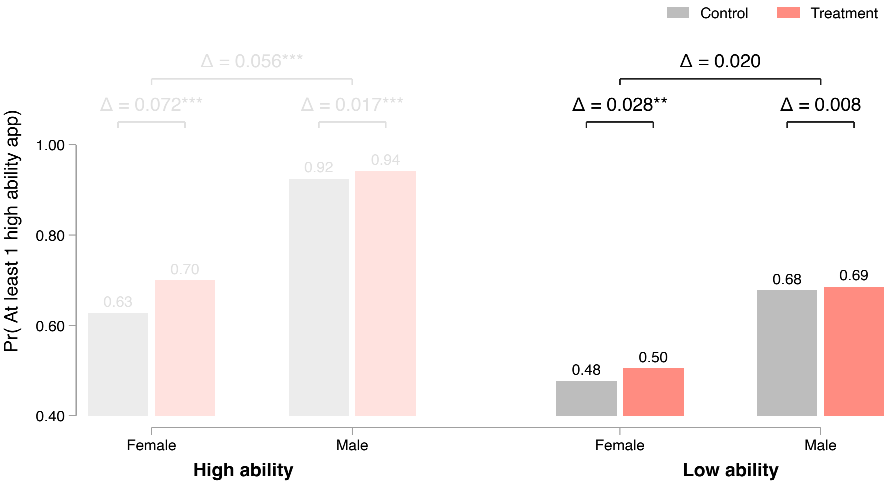

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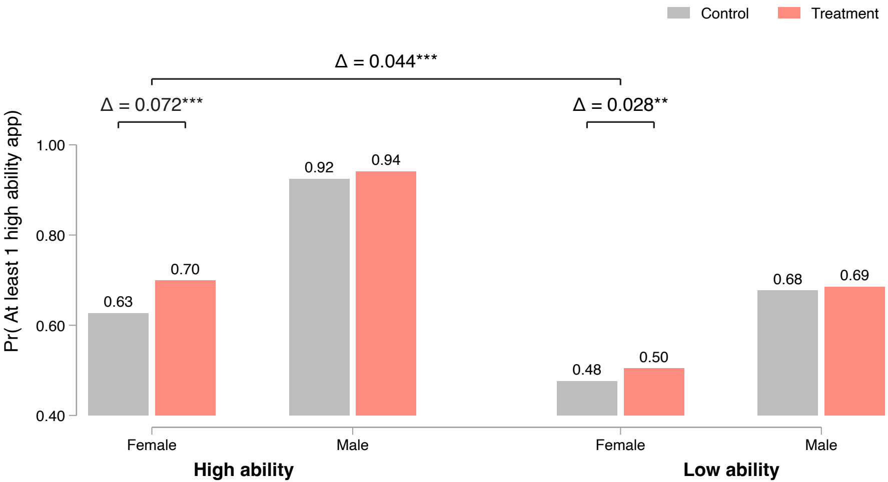
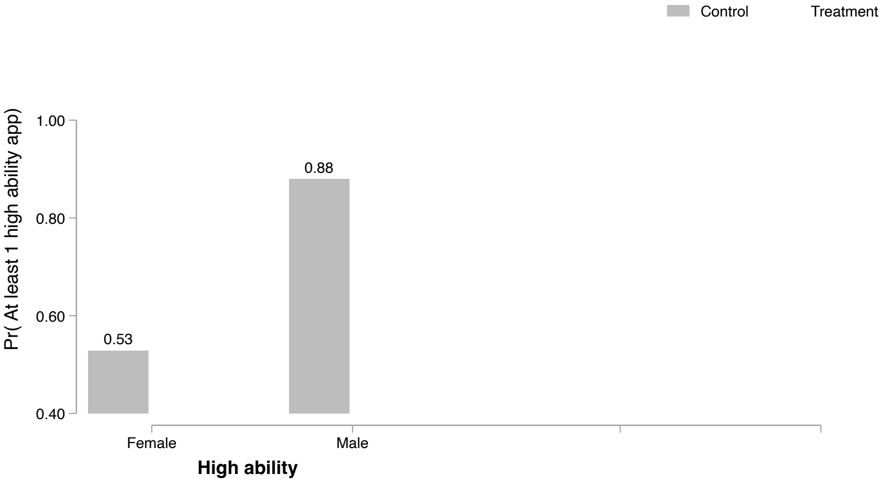


Back

SMALL FIRMS: Treatment draws in high ability women but effects smaller than for large firms



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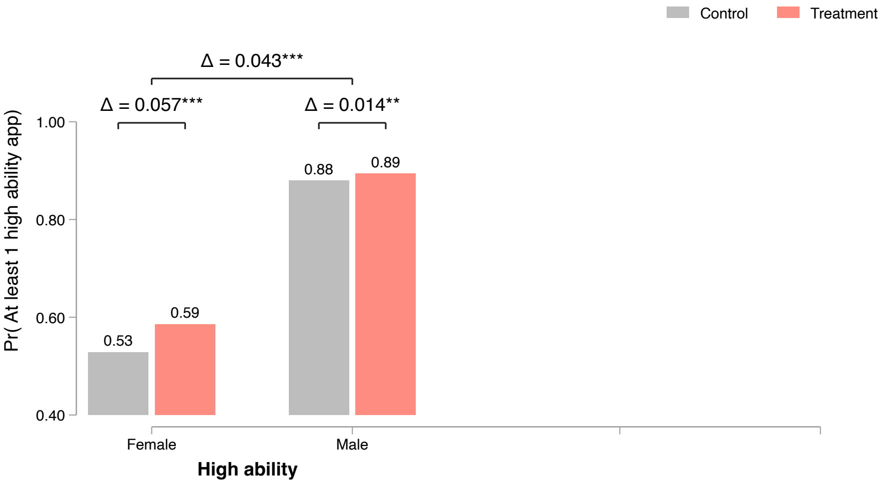
SMALL FIRMS: Treatment draws in high ability women but effects smaller than for large firms



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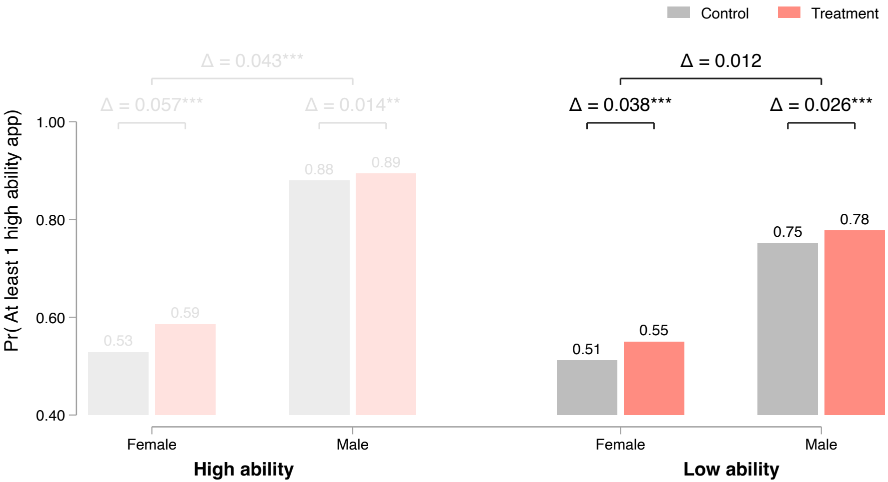
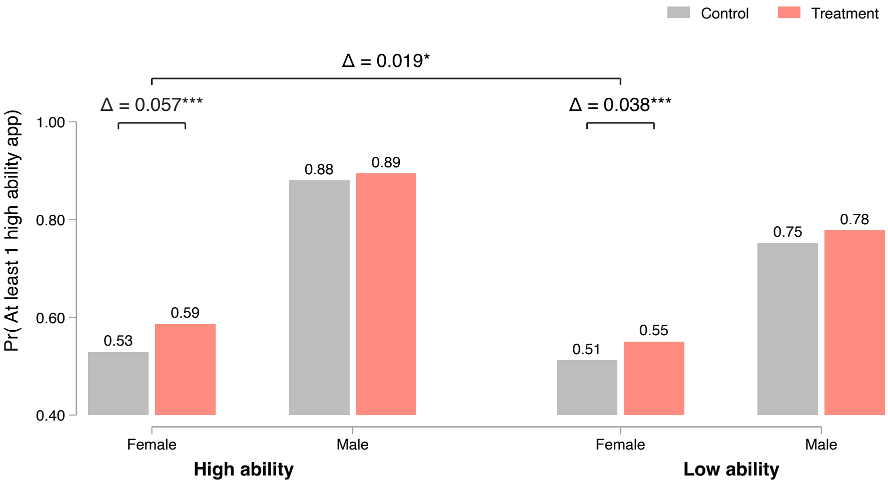
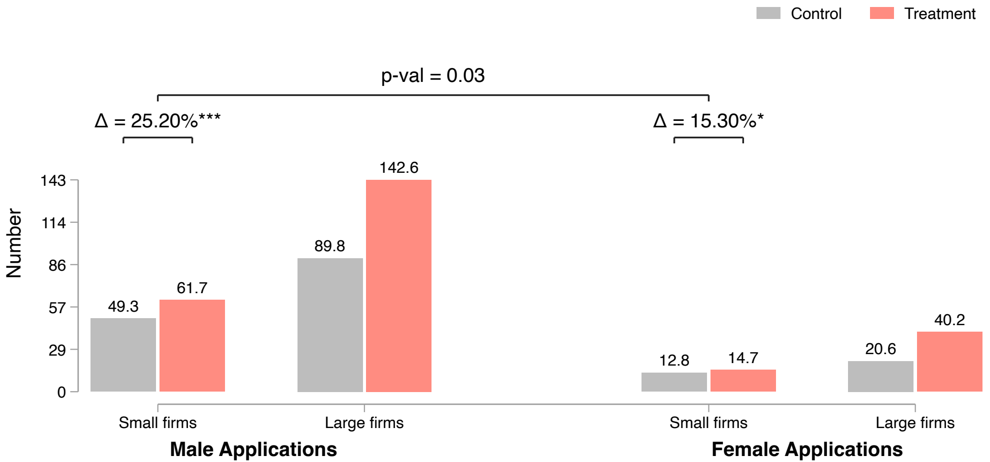


SMALL FIRMS: Treatment draws in high ability women but effects smaller than for large firms



Treatment has limited effect on applications to jobs from small firms



OLS table | Poisson table | IV | Back to main results

Identifying top candidates

- Goal: assess which candidates are the best across resume characteristics
- Identify top 3 values of GPA, experience, education, management and expected salary in applicant pool
- Flag all candidates who have these top values
- Construct share of the 5 fields an applicant is top in
- Top candidate: person with highest share of the 5 fields they are top in

Back

Data and Empirical Approach

- Results are reduced form
- OLS for log wages, match quality:

$$Y_j = \alpha + \beta T_j + \epsilon_j \tag{1}$$

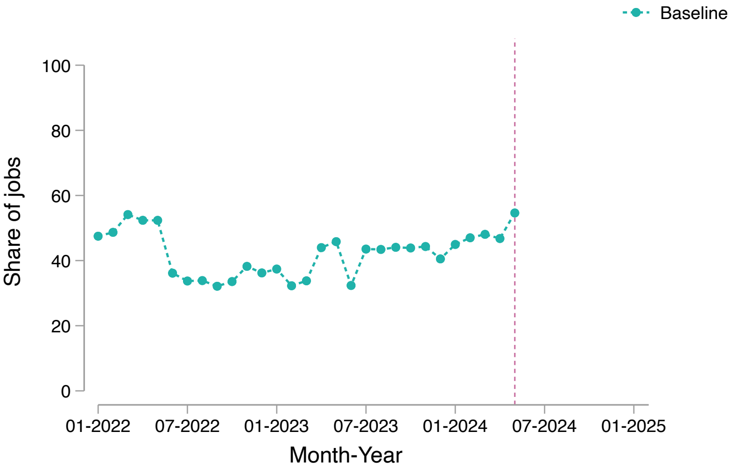
- Poisson for counts data (views, applications), reporting incident rate ratios, allows for interpretation in percents:

$$Y_j = exp(\alpha + \beta T_j) \tag{2}$$

Back

First stage: Share of jobs with visible salary

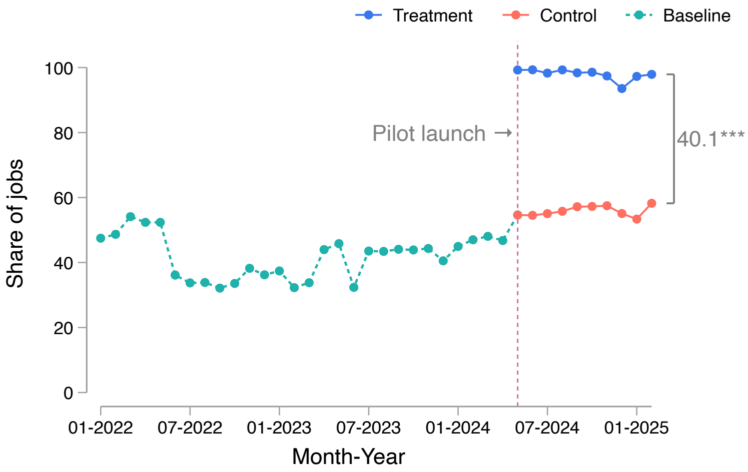
▸ Back



▸ Table ▸ No job or firm exits

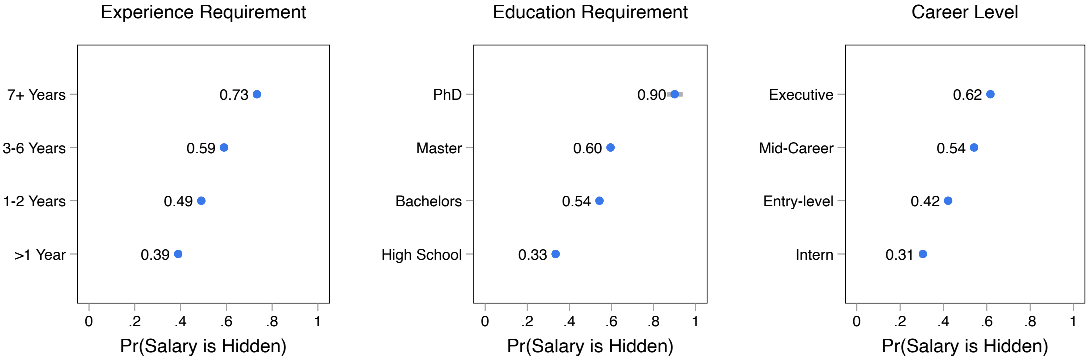
First stage: Share of jobs with visible salary increaes by 41 pp in treatment group

▸ Back



▸ Table ▸ No job or firm exits

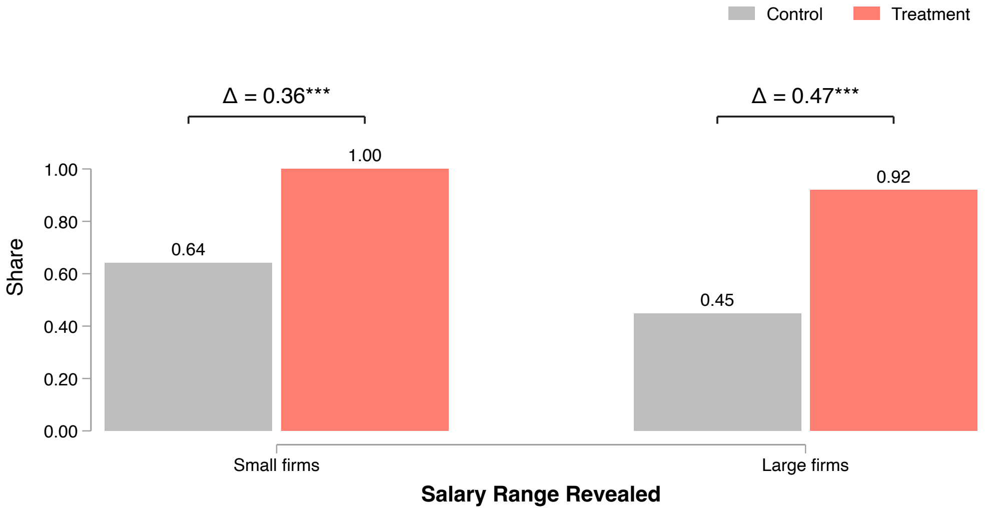
Fact 1: Jobs with hidden salary are more complex ▸ Back



Similar patterns documented in the U.S. and Chile (Batra et al., 2023; Banfi and Villena-Roldán, 2019) ▸ Details

The treatment induces them to reveal

Back



Treatment does not alter advertised salaries

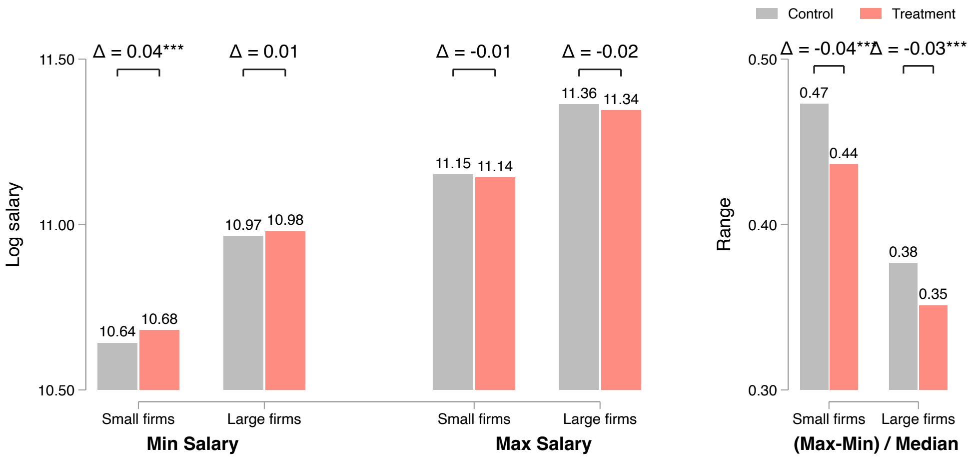
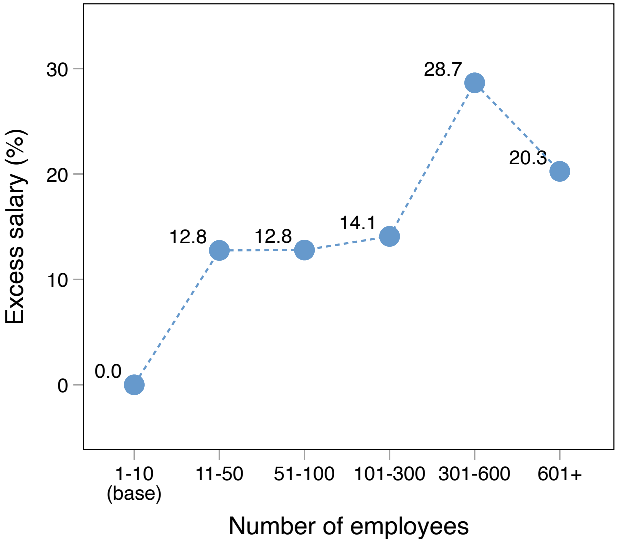


Table IV Back

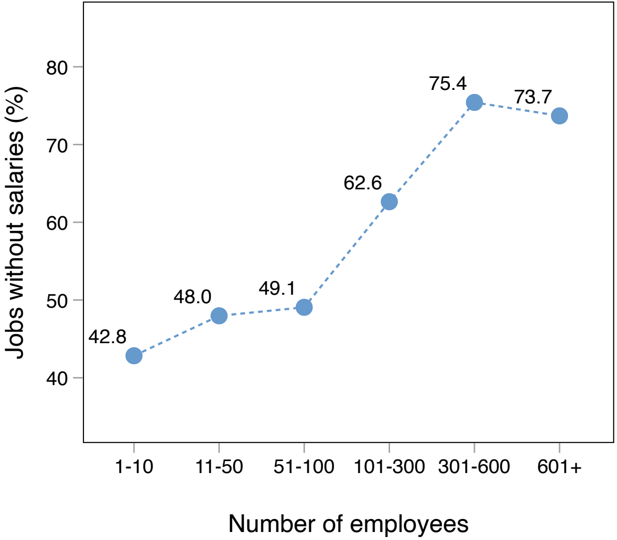
Large firms pay more conditional on job type

Model Fact 1



Large firms hide salaries more often

◂ Back ◂ Fact 1



Large firms offer fewer amenities preferred by women: Survey data

Model Fact 3

Table 1: Amenities and firm size – historical data

	Flex hours		Remote work		Need vehicle		Overnight travel		Transport support	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Large firm	-0.12***	-0.10***	-0.07***	-0.06***	0.03***	0.02***	0.05***	0.00	-0.01**	-0.01**
	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Small firm	0.45	0.45	0.19	0.19	0.07	0.07	0.21	0.21	0.23	0.23
Observations	35,382	35,382	35,382	35,382	35,382	35,382	35,382	35,382	35,382	35,382
Industry FE		✓		✓		✓		✓		✓
Occupation FE		✓		✓		✓		✓		✓
Education		✓		✓		✓		✓		✓
Career level		✓		✓		✓		✓		✓
Experience		✓		✓		✓		✓		✓

Non-wage characteristics fade in importance once wages are revealed

Model Back

	Control: Salary Hidden		Treatment	
	(1) N female apps	(2) N male apps	(3) N female apps	(4) N male apps
Large firm	1.13	1.28***	1.60***	1.42***
	(0.13)	(0.08)	(0.19)	(0.11)
Remote work	2.69***	1.71***	1.90***	1.57***
	(0.29)	(0.18)	(0.23)	(0.14)
Transport	1.18	1.02	0.56**	0.74**
	(0.24)	(0.21)	(0.13)	(0.11)
Flex work	1.03	0.89	0.66*	0.74***
	(0.16)	(0.11)	(0.14)	(0.08)
Safe environment	1.40	1.39*	1.02	1.01
	(0.31)	(0.27)	(0.17)	(0.12)
Equal opp. employer	1.73**	1.40**	1.27	1.24
	(0.39)	(0.22)	(0.31)	(0.19)
Dependent var. mean	18.31	72.07	25.52	95.98
Observations	3,851	3,851	10,972	10,972

Large firms are less likely to offer remote work

Panel A: Control jobs

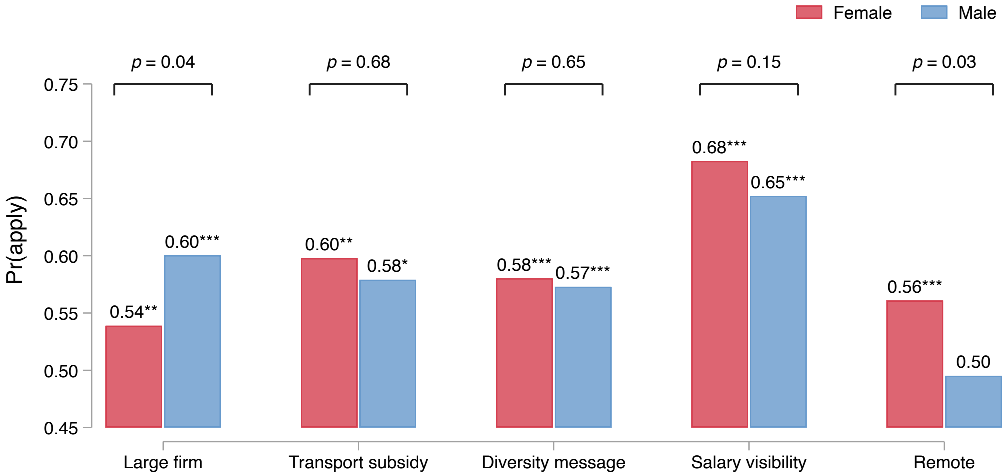
	(1) Remote work	(2) Transport	(3) Flex work	(4) Safe envir.	(5) Equal opp. employer
Large firm	-0.05***	0.00	-0.02***	0.01***	0.01***
	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)
Small firm	0.10	0.02	0.05	0.02	0.02
Observations	8,742	8,742	8,742	8,742	8,742

Panel B: Treatment jobs

	(1) Remote work	(2) Transport	(3) Flex work	(4) Safe envir.	(5) Equal opp. employer
Large firm	-0.05***	-0.00	-0.02***	0.01***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Small firm	0.10	0.02	0.05	0.02	0.02
Observations	10,972	10,972	10,972	10,972	10,972

Back

Preferences for job attributes



▸ Return

Women do not differentially misperceive competition in jobs with hidden pay

	Guess within 10% of female apps		Guess within 10% of male apps	
	(1) Small firm	(2) Large firm	(3) Small firm	(4) Large firm
Female	-0.015	-0.005	0.008	-0.009
	(0.019)	(0.018)	(0.018)	(0.018)
Male Mean	0.050	0.040	0.035	0.040
Observations	456	456	456	456

Back

Women do not differentially misperceive firm size in jobs with hidden pay

	Guess firm size correctly:	
	(1)	(2)
	Small firm	Large firm
Female	0.037	-0.017
	(0.049)	(0.045)
Male Mean	0.347	0.728
Observations	402	402

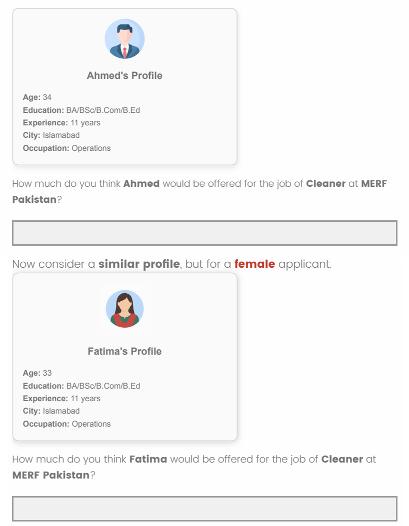
Back

Gender gaps in network size and composition do not drive access to pay info

	(1) Network size	(2) Share of network that provides job advice	(3) Share of network employed	(4) Share of employed network that earns more	(5) Share of workers' apps to treated jobs
Female	-0.050	-0.028	-0.088***	-0.012	0.073
	(0.647)	(0.031)	(0.029)	(0.032)	(0.099)
Female × Share of network employed					-0.109
					(0.123)
Share of network employed					0.038
					(0.096)
Male Mean	3.945	0.720	0.786	0.615	0.682
Observations	422	421	421	397	421

◀ Back

Perceived gender discrimination is not sizeable



Ahmed's Profile

Age: 34
Education: BA/BSc/B.Com/B.Ed
Experience: 11 years
City: Islamabad
Occupation: Operations

How much do you think **Ahmed** would be offered for the job of **Cleaner** at **MERF Pakistan**?

Now consider a **similar profile**, but for a **female** applicant.

Fatima's Profile

Age: 33
Education: BA/BSc/B.Com/B.Ed
Experience: 11 years
City: Islamabad
Occupation: Operations

How much do you think **Fatima** would be offered for the job of **Cleaner** at **MERF Pakistan**?

	(1) **Perceive discrimination**	(2) **Ratio: Own offer to Perceived max pay**
Female	0.054 (0.041)	0.086 (0.080)
Male Mean	0.224	0.908
Observations	444	444

Back

No gender differences in ambiguity aversion

◂ Back

🎮 **Let's play a game:**

Pick the Right Ball to Win PKR 500!

A **ball** will be **randomly drawn** by our algorithm.
You **choose which bag** the ball is drawn from.
If the right color ball is drawn, you win a **cash prize**.

Pick one of these bags 👇

💰 **Bag One – You know what's inside**

🔴🔴🔴🔴 = **4 RED balls**
🟡🟡🟡🟡🟡🟡 = **6 YELLOW balls**

🎯 **You win if:** **A RED ball is drawn**

🎲 **Chance of winning:** 4 out of 10 = **40%**

BOTTOM LINE: You know what's in the bag, but can't choose the color that wins.

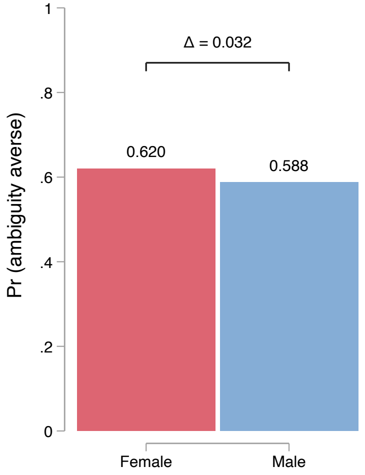
💰 **Bag Two – Mystery Bag**

🔴 ❓ 🟡 ❓ **You don't know** how many RED or YELLOW balls are inside.

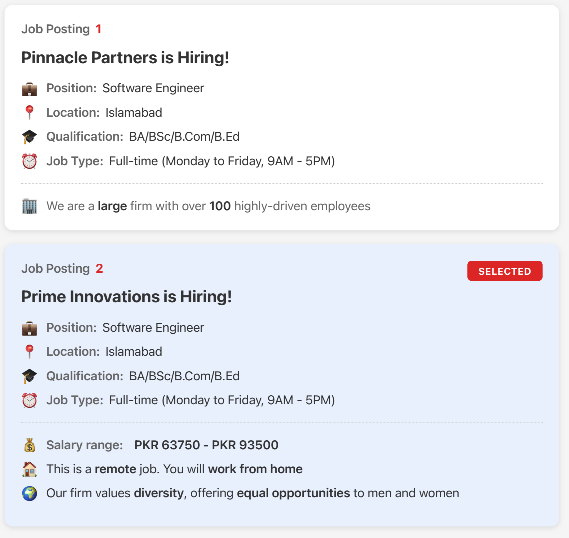
❗ But you **choose which color counts as a win** — RED or YELLOW.

🎯 **You win if:** The ball that's drawn matches the **color you chose**
🎲 **Chance of winning:** *Unknown*

BOTTOM LINE: You choose the winning color, but don't know what's in the bag.



Discrete-choice experiment: design



- 4 discrete choices, 8 options
- In each ad 50% likelihood:
  - Salary disclosed
  - Large firm
  - Diversity
  - Transport subsidy
- If salary disclosed
  - Min $\in \{0.7, 0.75, 0.8\}$
  - Max $\in \{1.1, 1.15, 1.2\}$

  of own salary

◂ Back to remote pref. ◂ Back to wage pref.
◂ Back to transparency pref. ◂ Back to model ▸ Specification

Discrete-choice experiment: Econometric specification ▸ Design

Conditional logit: Choice probability that respondent *i* chooses job *j* in scenario *c*:

$$P_{ijc} = \frac{\exp(\beta^\top X_{jc})}{\sum_{\ell \in C_{ic}} \exp(\beta^\top X_{\ell c})}$$

- C_{ic} = set of jobs offered to respondent *i* in scenario *c*
- X_{jc} = attributes for job *j* in scenario *c*: [Transport, Large firm, Inclusive language, Salary visible, Remote]
- β = vector of preference parameters
- Respondent-choice FE and errors clustered at respondent level

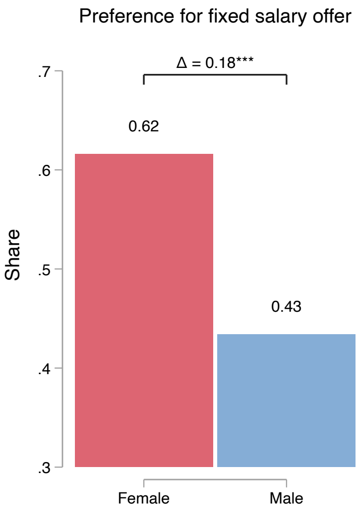
Average marginal effect:

$$\text{AME}_k = \frac{1}{N} \sum_i \sum_c \sum_{j \in C_{ic}} \Big[P_{ijc}\big(X_{jc}^{(k)} = 1\big) - P_{ijc}\big(X_{jc}^{(k)} = 0\big) \Big]$$

- $X_{jc}^{(k)}$ = value of attribute *k* for job *j* in scenario *c*
- $N = \sum_i \sum_c |C_{ic}|$ = total number of job profiles shown
- Interpretation: average p.p. Δ in probability of job choice when *k* switches $0 \to 1$, all else equal

Most women prefer a fixed salary offer

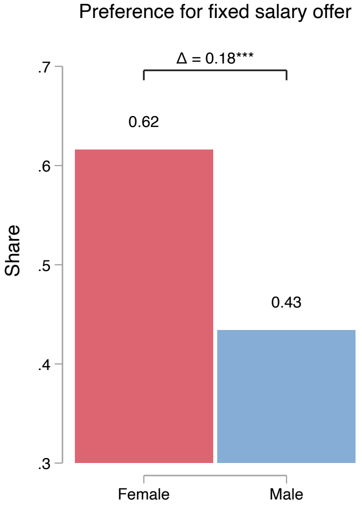
Mechanisms

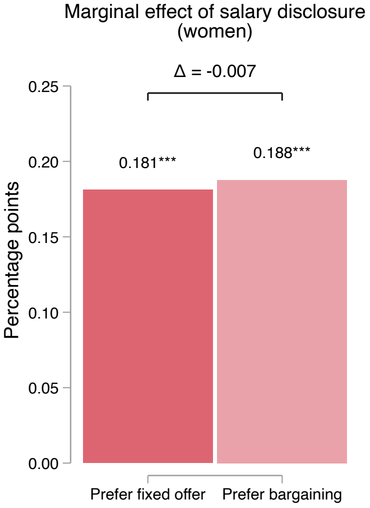


> *Imagine you have two job offers for the same position.*
>
> *One firm sends you a final salary offer.*
>
> *The other invites you to negotiate salary.*
>
> *Neither listed a salary in the job ad.*
>
> *You can only respond to one firm.*
>
> *Which would you respond to?*

...but taste for transparency does not differ by bargaining aversion

Mechanisms





Female hires in treated vs. control jobs

	N. Women			N. Men		
	(1) All firms	(2) Large firms	(3) Small firms	(4) All firms	(5) Large firms	(6) Small firms
Treated Job	-0.01	-0.01	-0.00	0.05	-0.04	0.08
	(0.03)	(0.03)	(0.04)	(0.06)	(0.09)	(0.07)
Control Job Mean	0.49	0.32	0.55	1.36	1.13	1.44
Observations	5,685	1,471	4,214	5,685	1,471	4,214

Back

Treatment effects on salaries

	(1) **Log Max Salary**	(2) **Log Min Salary**	(3) **Salary Range: (Max-Min)/Median**
Treated Job	-0.01	0.03***	-0.03***
	(0.01)	(0.01)	(0.00)
Constant	11.24	10.77	0.43
Observations	19,890	19,890	19,890

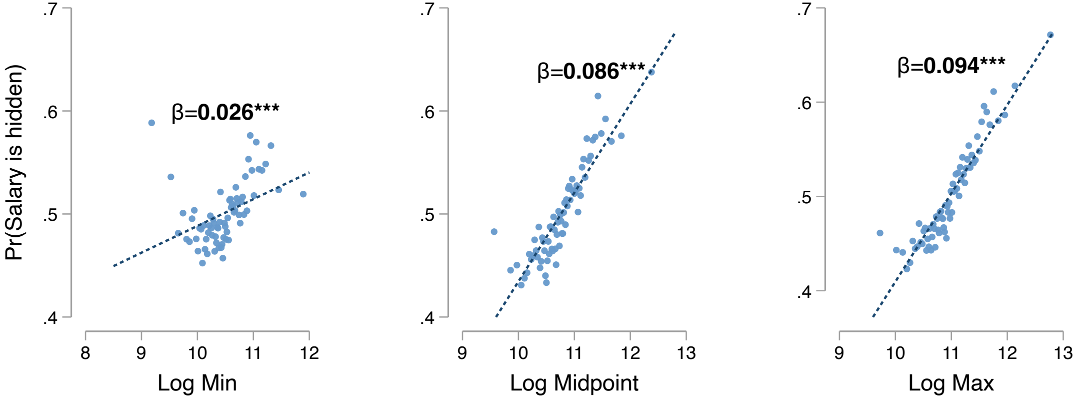
Back

Salary: 3% higher min; more compact range

IV OLS table Back to HTE by firm size

Hidden salaries are higher for otherwise similar jobs

◂ Back to model ◂ Back to large firm salaries



Why don't workers infer wages from non-wage attributes?

Firm size

- Empirically, large firms pay more and hide more $\Rightarrow$ Higher salaries more likely to be hidden ▸ Evidence
- But only 11% of surveyed workers think hidden salaries are higher

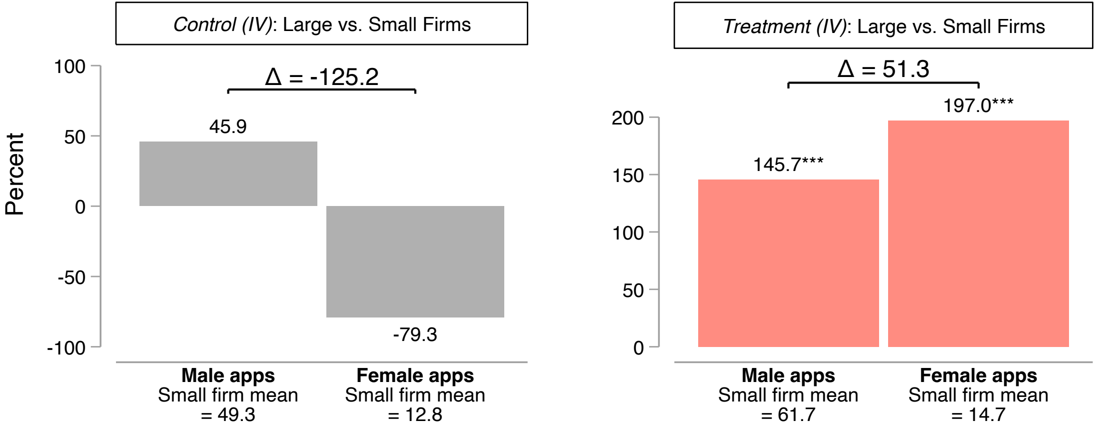
  $\hookrightarrow$ Either don't know that large firms pay more or don't apply the correlation
- Plausible to not know correlation because
  - When large firms post, it is for low-paid jobs
  - Wage dispersion is highest in large firms

Amenities

- The signal is misleading. Often, empirically $Corr(w, a) > 0$ due to omitted variable bias
- Applying correlation implies lower w $\rightarrow$ opacity reduces utility

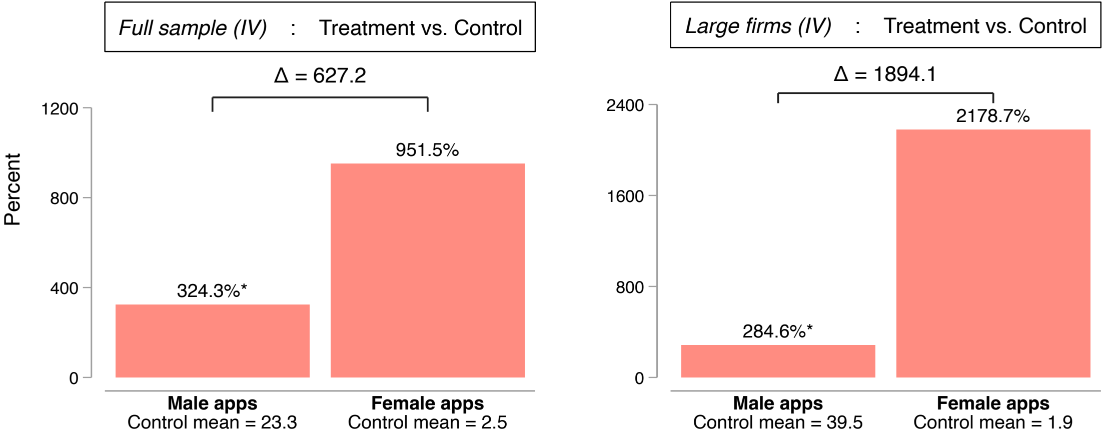
◂ Back

IV: Gender gaps in directed search to large firms reversed



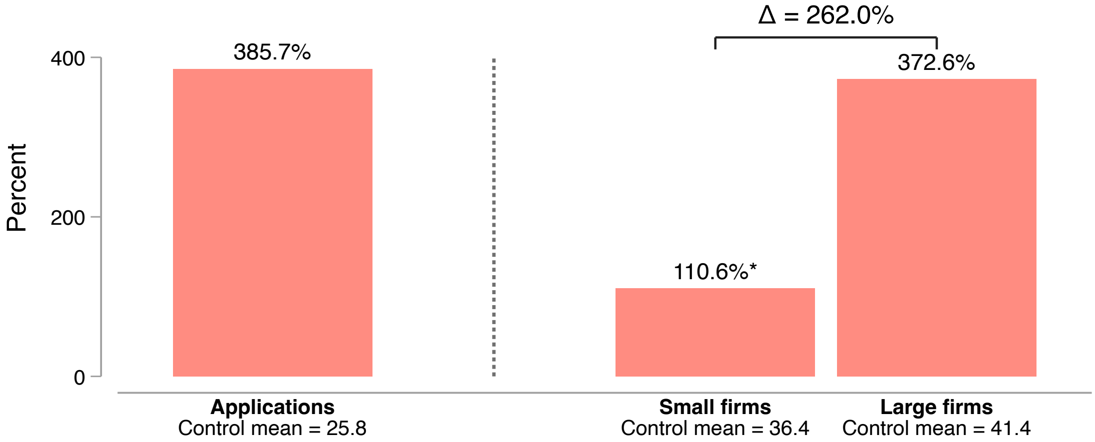
Back to reduced form

IV: Transparency impacts larger on female apps, especially to large firms



Reduced form

IV: Transparency raises applications, especially to large firms



Back to reduced form

Saturation design: More 'treated' Workers direct search to higher paying jobs

Share Apps to Treated Jobs with Above Median...	Min Salary Premium		Median Salary Premium		Max Salary Premium	
	(1) Male	(2) Female	(3) Male	(4) Female	(5) Male	(6) Female
High Saturation Market	0.05**	0.05**	0.05***	0.04**	0.03***	0.02**
	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)	(0.01)
Low Saturation Market Mean	0.38	0.42	0.38	0.41	0.32	0.32
Observations	117,080	33,502	117,080	33,502	117,080	33,502

Notes: Errors clustered at market level. Salary premium calculated by residualizing on job characteristics.

◂ Back

Does salary disclosure increase the probability a woman is hired at large firms? Back

$$\Pr(\text{hired})_g = \Pr(\text{applies})_g \times \Pr(\text{hired} \mid \text{applied})_g$$

Does salary disclosure increase the probability a woman is hired at large firms?

Back

$$\Pr(\text{hired})_g = \Pr(\text{applies})_g \times \Pr(\text{hired} \mid \text{applied})_g$$

Supply side: N_g = N. men or women in labor market, A_g = N. male or female applicants to a job:

$$A_g = N_G \times \Pr(\text{applies})_g \quad \Rightarrow \quad \Pr(\text{applies})_g = \frac{A_G}{N_G}$$

Does salary disclosure increase the probability a woman is hired at large firms? Back

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$$A_g = N_G \times \Pr(\text{applies})_g \quad \Rightarrow \quad \Pr(\text{applies})_g = \frac{A_G}{N_G}$$

Demand side: Assume random matching; H_g = N. male or female hires, T = total applicants:

$$\Pr(\text{hired} \mid \text{applied})_g = \frac{H_g}{T}$$

Estimating Treatment Effect

Back

Combining both terms:

$$\Pr(\text{hired})_g \approx \frac{A_g}{N_g} \cdot \frac{H_g}{T}$$

Estimating Treatment Effect

Back

Combining both terms:

$$\Pr(\text{hired})_g \approx \frac{A_g}{N_g} \cdot \frac{H_g}{T}$$

Treatment effect:

$$R_g = \frac{\Pr(\text{hired} \mid \text{treatment})_g}{\Pr(\text{hired} \mid \text{control})_g} = \frac{(A'_g/T') \cdot H'_g}{(A_g/T) \cdot H_g}$$

Since N_g is constant across groups, it cancels in ratios.

Estimating Treatment Effect ◂ Back

Combining both terms:

$$\Pr(\text{hired})_g \approx \frac{A_g}{N_g} \cdot \frac{H_g}{T}$$

Treatment effect:

$$R_g = \frac{\Pr(\text{hired} \mid \text{treatment})_g}{\Pr(\text{hired} \mid \text{control})_g} = \frac{(A'_g/T') \cdot H'_g}{(A_g/T) \cdot H_g}$$

Since N_g is constant across groups, it cancels in ratios.

Empirically, $H_g \approx H'_g \Rightarrow$

$$R_g = \frac{A'_g/T'}{A_g/T}$$

This gives the proportional change in a woman's hiring probability due to disclosure.

Calculation ◂ Back

Women. Using observed job-level means:

$$A_f = 20.613, \qquad T = 110.448, \qquad \Rightarrow \frac{A_f}{T} = 0.1867$$

$$A_f' = 40.239, \qquad T' = 182.912, \qquad \Rightarrow \frac{A_f'}{T'} = 0.2200$$

$$R_f = \frac{0.2200}{0.1867} = \boxed{1.178}$$

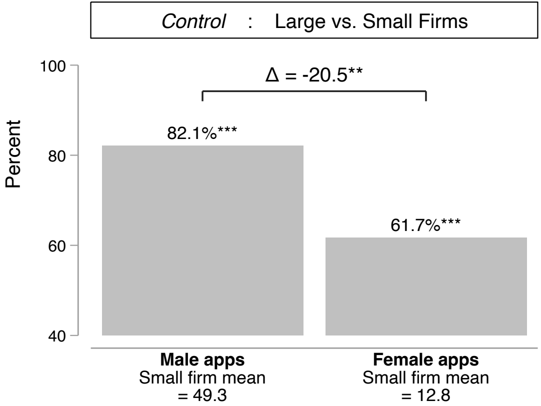
Men. Using observed job-level means:

$$A_m = 89.789, \qquad T = 110.448, \qquad \Rightarrow \frac{A_m}{T} = 0.8131$$

$$A_m' = 142.581, \qquad T' = 182.912, \qquad \Rightarrow \frac{A_m'}{T'} = 0.7795$$

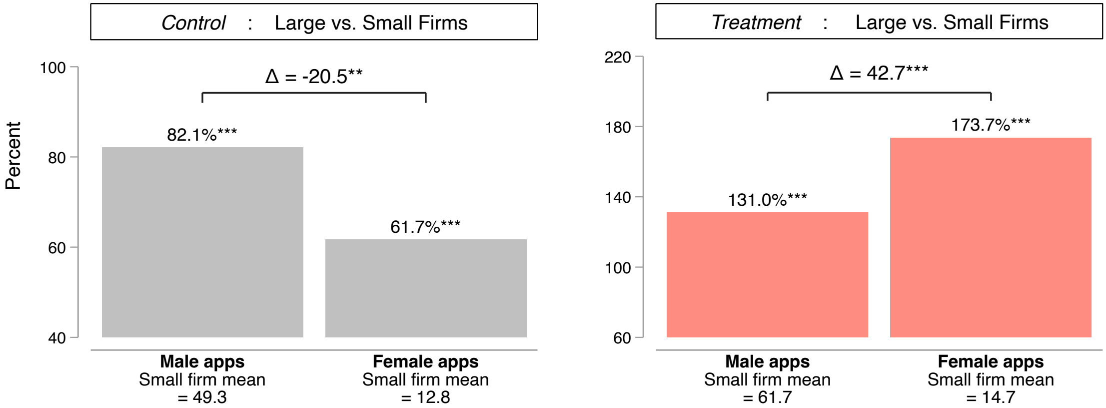
$$R_m = \frac{0.7795}{0.8131} = \boxed{0.9587}$$

Gender gaps in directed search to large firms reversed



▸ OLS table ▸ Poisson table ▸ IV graph ▸ Back

Gender gaps in directed search to large firms reversed



OLS table | Poisson table | IV graph | Back

Men and women value salary transparency similarly

Design | Model | Mechanisms

Discrete-choice experiment with 438 workers

Job Posting **1**

Pinnacle Partners is Hiring!

- **Position:** Software Engineer
- **Location:** Islamabad
- **Qualification:** BA/BSc/B.Com/B.Ed
- **Job Type:** Full-time (Monday to Friday, 9AM - 5PM)

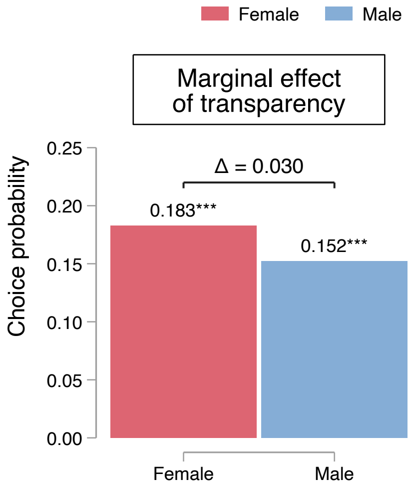
We are a **large** firm with over **100** highly-driven employees

Job Posting **2** SELECTED

Prime Innovations is Hiring!

- **Position:** Software Engineer
- **Location:** Islamabad
- **Qualification:** BA/BSc/B.Com/B.Ed
- **Job Type:** Full-time (Monday to Friday, 9AM - 5PM)

- **Salary range:** **PKR 63750 - PKR 93500**
- This is a **remote** job. You will **work from home**
- Our firm values **diversity**, offering **equal opportunities** to men and women



Men and women value higher wages similarly

▸ Design ◂ Model ◂ Mechanisms

Discrete-choice experiment with 438 workers

Job Posting **1**

Pinnacle Partners is Hiring!

💼 Position: Software Engineer

📍 Location: Islamabad

🎓 Qualification: BA/BSc/B.Com/B.Ed

⏰ Job Type: Full-time (Monday to Friday, 9AM - 5PM)

💰 Salary range: **PKR 90750 - PKR 100500**

Job Posting **2** SELECTED

Prime Innovations is Hiring!

💼 Position: Software Engineer

📍 Location: Islamabad

🎓 Qualification: BA/BSc/B.Com/B.Ed

⏰ Job Type: Full-time (Monday to Friday, 9AM - 5PM)

💰 Salary range: **PKR 63750 - PKR 93500**

🏠 This is a **remote** job. You will **work from home**

🌍 Our firm values **diversity**, offering **equal opportunities** to men and women